Conversational systems

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Outline

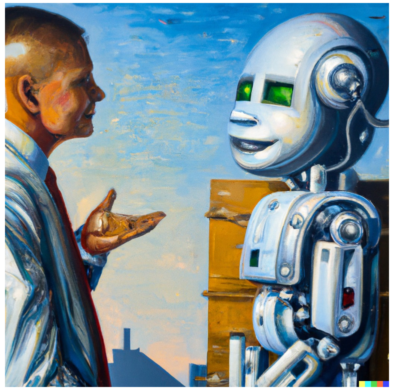
- Theoretical Background
- Approaches
- Data and evaluation
- Applications and ethical issues

Definition (conversational systems)

- Conversational systems:
  - human-machine dialogue
- Domain:
  - Open domain
    - Ex IBM Watson Jeopardy : mimicking Jeopardy! TV game =
      - the presenter utters an answer
        - “He is known for his horror novels, including “Carrie” and “The Shining.”
      - the candidates have to find the related question : ?
        - “Who is Stephen King?”
  - Closed domain (train ticket reservation)
    - Ex : I want to go from Paris to Marseille tomorrow around 9 am.

Definition (conversational systems)

- Communication situation:
  - Face-to-face, vocal, written



Generated by chatGPT

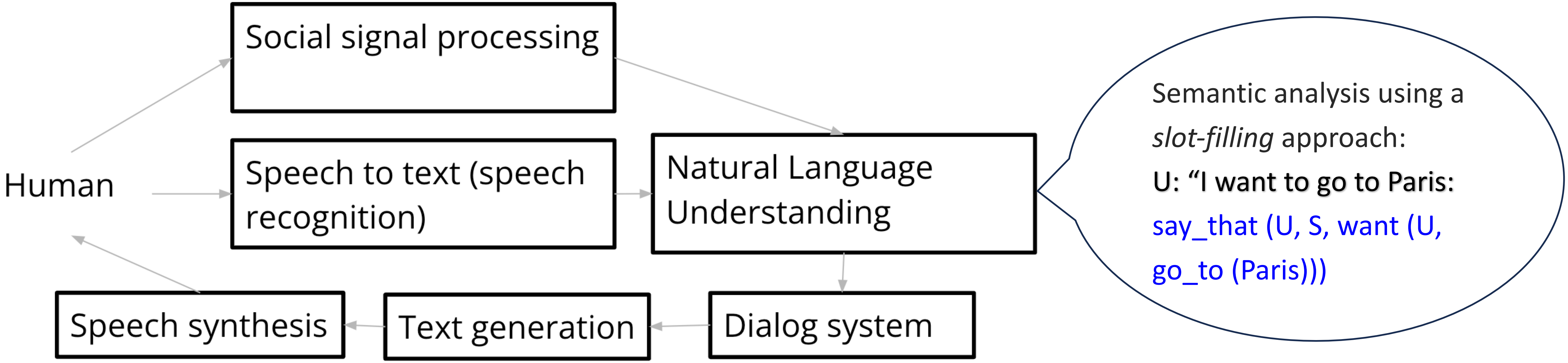
  - One-shot or longitudinal (more than one interaction with a same agent)

Task-oriented systems

- Complete a given task for the user
- Could be a question answering problem
  - where the task is to give information to the user
- In a specific application domain, ex:
  - a train travel reservation,
  - customer relationship management,
  - movie recommendations, …
- Or for open-domain question answering

Task-oriented systems

- 1st well-known task-oriented systems: Trains (Ferguson, 1996)
  - Task: booking train travel
  - Vidéo : https://www.georgeferguson.com/george/projects/trains-session.html
  - Speech-based dialogue : practice – what are the modules required here?



“You can take the following train lines” + *display*

Task-oriented systems – slot filling approaches

- The system asks **the information required to fill in a form**, and provides the user with the result of the request
  - Ex: Ask the user for a departure city, a destination city, a time, and whether the trip is round-trip or not
- **System Initiative** = the form is filled in step by step
  - The user *must* comply, and provide the information step by step, in the right order
- **Mixed Initiative** = information is extracted from the user's input and used to fill in the form
  - The user can provide information more freely

Social-oriented systems
(focus on social interactions)

- where the primary goal is to engage the user
- Focus on the socio-emotional layer of the interaction.
- 1st well-known social-oriented system: **Eliza system** (Weizenbaum, 1966)
  - Written interactions
  - Following Rogerian psychology interview (from the psychologist Carl Rogers):
    - Principle :
      - person-centered therapy,
      - reflecting patients’ statements back at them
      - to encourage the patients to express themselves.

Social-oriented systems
(focus on social interactions)

Part of a conversation between a user and a machine (**Eliza system**):

U : He says I'm depressed much of the time.

M : I AM SORRY TO HEAR YOU ARE DEPRESSED

U : It's true. I am unhappy

M : DO YOU THINK COMING HERE WILL HELP YOU NOT TO BE UNHAPPY

U : I need some help, that much seems certain.

M : WHAT WOULD IT MEAN TO YOU IF YOU GOT SOME HELP

U : Perhaps I could learn to get along with my mother.

M : TELL ME MORE ABOUT YOUR FAMILY

U : My mother takes care of me.

M : WHO ELSE IN YOUR FAMILY TAKES CARE OF YOU

U : My father.

Parenthesis: Ethical thoughts

- What do you mean about this kind of conversational systems (psychological agents) ?
- The risks : social irrelevance of the automatic answers, manipulation, attachment

A reference : https://www.cnrs.fr/en/update/public-research-has-play-key-watchdog-role-use-social-robots

Novel with ELIZA: *Un tout petit monde* de David Lodge

Eliza system: how does it work?

- the user input is read and inspected for the presence of a keyword.
- the sentence is transformed according to a rule associated with the keyword
  - ***Decomposition rules*** are used to represent the input sentence
    - Ex: (1) It seems that (2) you (3) hate (4) me -> (0 YOU 0 ME)
  - Responses are generated by ***reassembly rules*** associated with selected decomposition rules
    - Ex: WHAT MAKES YOU THINK I (3) YOU
- The text so computed or retrieved is then printed out.

From https://redirect.cs.umbc.edu/courses/331/papers/eliza.html

No borderline between task and social

- First task-oriented systems ignored questions of social suitability,
  - Smiling in the face of a user’s frustration
- There is an impact of social bonds on task performance (Dörnyei and Kormos, 2000; Kamdar and Van Dyne, 2007).
  - Children learn better when paired with friends

No borderline between task and social

Example of conversational systems mixing task and social

- Integrate conversational behaviors dedicated to strengthening social bounds with the user with the view to improving effectiveness for
  - A museum guide Bickmore et al. (2013)
  - A movie recommendation agent, Lee and Choi (2017)

Is chatGPT mixing task and social? Yes :

- it focuses on the safety of the answer in order to avoid inappropriate answers and social norm (Thoppilan et al., 2022).
- It develops social strategies to make the user like the interaction (ex: sycophancy)

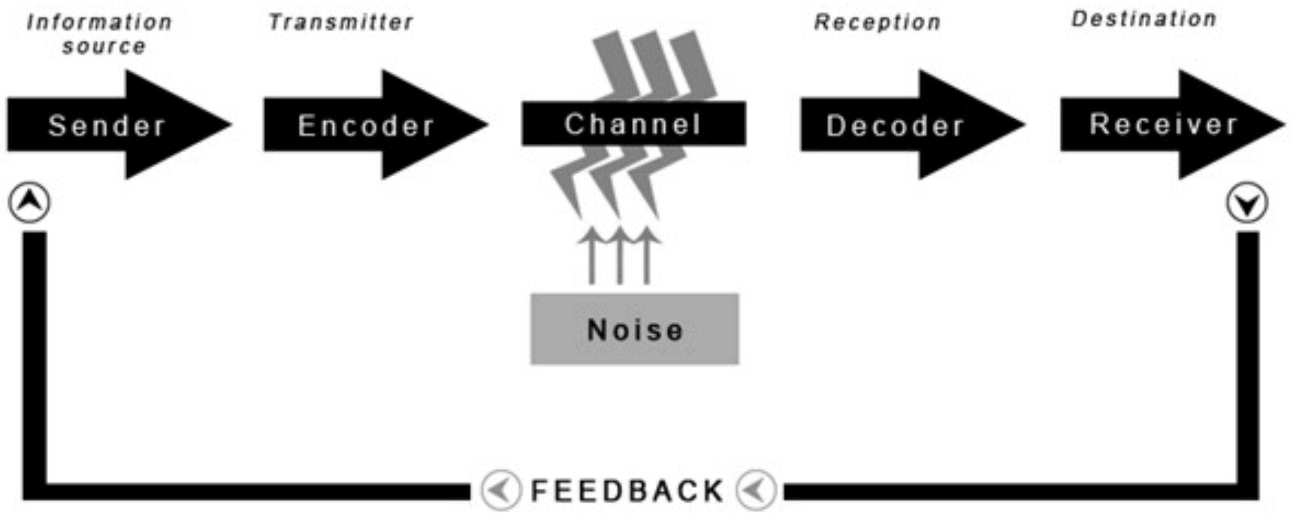
Definitions (human communication)

- Building conversational systems requires an in-depth understanding of human communication to obtain *natural dialogue in natural language*
  - to study natural *language* we need to analyze lexicon, syntax, prosody, semantics, pragmatics (see lecture on sentiment analysis)
  - To study natural *dialogue* we need to analyze human dialog structure cohesion, coherence
- Which implies lot of links with several research disciplines: linguistics, pragmatics[1], cognitive sciences, social sciences

1. the branch of linguistics dealing with language in use and the contexts in which it is used, including such matters as deixis, the taking of turns in conversation, text organization, presupposition, and implicature.

Some examples of different theories used for the design of conversational systems

- Former model: Mathematical **Theory of Communication** [Shannon, 1948] whereby a message is passed from sender to receiver via a channel.



Shannon-Weaver's Model of Communication

Now replaced by models that better integrate the collaborative nature of a conversation

Some examples of different theories used for the design of conversational systems

1/Example of linguistic models

- Models relying on “**grounding” concept** (Clark, 1996):
  - model the joint responsibility of the participants to ensure that contributions to the conversation are **mutually understood**

Example of collaborative process from the Switchboard corpus

```
S1: And I like light meals. Not all restaurants have light meals on their
menu
(...)
S2: Oh. Light menu, you mean like cooked light or,
S1: L I T E, like, uh, uh, things that aren't as heavy a meal.
S2: Oh, I see.
(...)
S1: that, you get more, ug fruit and maybe cottage cheese or a chicken
breast or things that aren't as, uh, they're not breaded and they're not
fried and they're not,
S2: Not a log of gravy and things like that
S1: Yeah.
```

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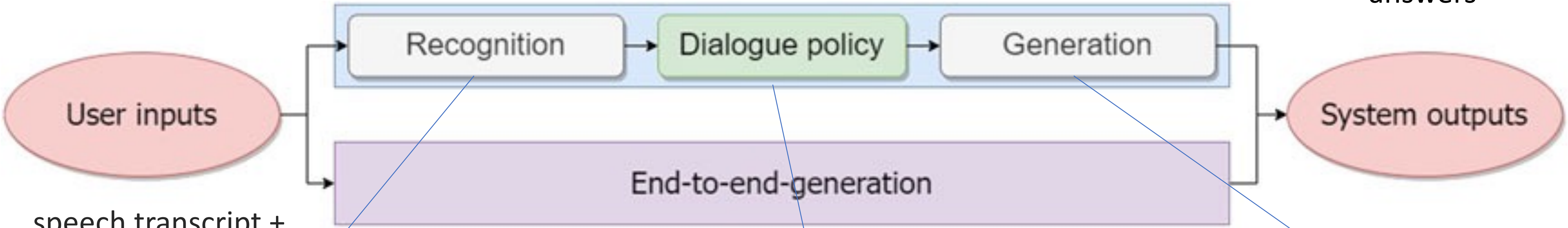
How does it work?

Clavel, C., Labeau, M., & Cassell, J. (2023). Socio-conversational Systems: Three Challenges at the Crossroads of Fields. Frontiers in Robotics and AI

- **Modular** vs. end-to-end approaches

Socially-relevant answers



speech transcript + visible/audible behaviors

Natural Language Understanding
Assigning the user input to labels
(ex: an intent - *book a train*
or an emotion - *frustration*)

Selecting the type of answer
(ex. an intent
- a suggestion of a particular train
or a suitable socio-emotional strategy
– empathy)

Generating the text and visible/audible behaviors corresponding to the selected dialogue policy

How does it work?

- Modular vs. **end-to-end approaches**

Socially-relevant
answers

speech transcript +
visible/audible behaviors

Directly generate the relevant answer
according to user inputs
Generation of the answer done *via* a (large)
language model with or without prompting

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  - focus on modular approaches
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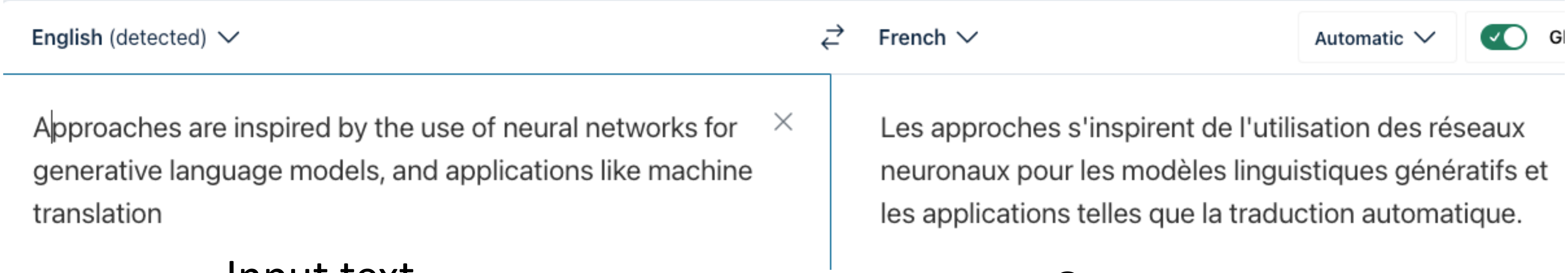
End-to-end approaches using machine translation paradigm



Input text

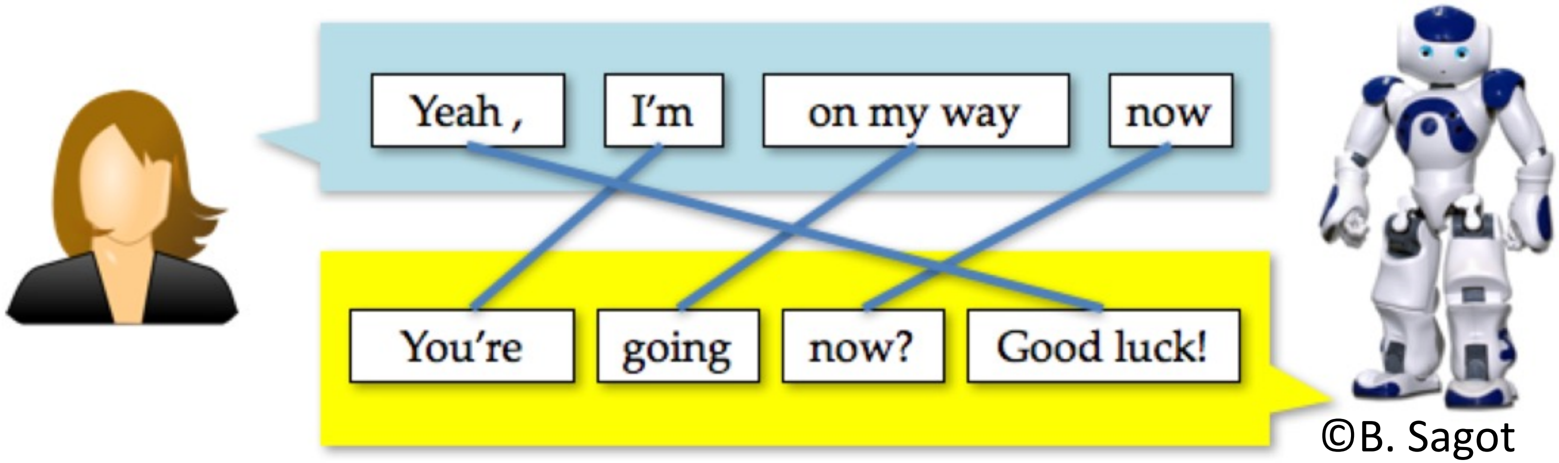
Output text

Dialogue Context:
A: I really need to start eating healthier.
B: I have to start eating better too.
A: What kind of food do you usually eat?
B: I try my best to eat only fruits, vegetables, and chicken.
A: Is that really all that you eat?
B: That’s basically it.
A: How do you stick to only those foods?

DialoGPT:
I eat a lot of fruit and veggies. I stick to a lot of things. I don’t eat a lot of junk food.

End-to-end approaches using machine translation paradigm

- Underlying paradigm: modelling answer generation as the task of generating a **transformation** of the user utterance (Ritter et al., 2010)



Exploit high-frequency patterns with phrase-based MT

"I am" → "you are"   "sick" → "get better"   "lovely!" → "thanks!"

End-to-End Approaches

Starting point: Serban et al. (2016) proposed one of the first end-to-end neural system dedicated to dialog, based on **recurrent neural networks**.

Now current models are based on **transformer architectures** trained on

- **question/answers pairs or,**
- **conversations for next utterance generation**

Formalization of the training task

- Each dialogue $\mathrm{d} = \{\mathrm{u_1}, \mathrm{u_2}, \mathrm{u_3} \dots \mathrm{u_m}\}$ consists of m turns
  - with $\mathrm{u_i}$ representing a specific turn
  - $\mathrm{u_i}$ is the vector $(\mathrm{x_1}, \mathrm{x_2}, .., \mathrm{x_N})$ representing the turn (N=size of the representation space)
- $h_i = \{u_{max(1,i-\omega)}, u_{i-\omega+1}, \dots u_i\}$ dialogue history (of fixed window size (ω))

Objective: Directly generate a response u_{i+1} as a function of the preceding sequence of utterances h_i in a dialogue

- Training objective:
  - Training data: set of human-human **unlabelled** dialogues $\mathrm{D} = \{\mathrm{d_1}, \mathrm{d_2}, \mathrm{d_3}, \dots \mathrm{d_n}\}$
  - find a model, denoted M, that will provide the next utterance u_{i+1} that is the closest to the human utterance based on the dialogue history h_i

End-to-End approaches for question answering

- Train a model on question/answer pairs
  - Dialog history h_i : just one question
  - Open/specific domain question-answer pairs
  - Possibility to introduce domain knowledge
    - Using knowledge bases
    - Last trend : Retrieval Augmented Generation (RAG) (Siriwardhana et al., 2023)
      - RAG can extract knowledge from an external knowledge base
      - merges two stages into one architecture
        - the retriever (selects passages relevant to a given question)
        - the reader (generates the answers from selected passages)
      - Can then be used with prompted llms

E2E approaches for dialogue systems: training corpora

Pre-trained transformers fine-tuned on conversations (dialog history, next utterance)

- **DialoGPT** (Zhang et al., 2019)
  - GPT2 trained on 147M conversation-like exchanges extracted from **Reddit**
- **BlenderBot** [Smith et al., 2020]
  - Pretrained on 1.5B **Reddit** comments and trained on various chitchat datasets
- **Lamda** (Language Models for Dialog Applications), [Thoppilan et al., 2022].
  - Pre-trained on dialog data from **public dialog data (**1.12B dialogs with 13.39B utterances) and other **public web documents** (2.97B documents)
- **GODEL** [Peng et al., 2022]
  - trained on web text data, and further trained on 551M **Reddit threads** and 5M instructions and knowledge-grounded dialogue datasets.
- **Koala and Vicuna** (Touvron et al., 2023)
  - finetuned LLaMA using **dialogue data from the web (ShareGPT**: sharing chatGPT conversations)

Etc. ….

Transformer for generative large language models

- **Generative** Pre-training Objective:
    - predict the next token from a long previous context (ex: 2048 tokens in GPT3)
- trained on a lot of data
    - ex: 300 billions tokens : Wikipedia, books corpora, and crawled webpages (CommonCrawl, WebText2) in GPT3
- Huge dimensional vectors
    - Ex: 175 billions parameters, in GPT 3 (vs. 150 billion for GPT-2)
- Can be used with prompts to obtain the next utterance

Prompting LLM to obtain next utterance

Example in a zero-shot setting

Voici la transcription d'un appel téléphonique entre un agent en réparation automobile et un client :
CLIENT: Oui bonjour j'aimerais savoir ça serait j'ai fissuré le pare-brise de ma voiture donc j'aimerais savoir si c'est possible de pour réparer
AGENT: Oui c'est une fissure qui dépasse les quatre centimètres madame ou c'est inférieur ?
CLIENT: Ah oui il y a quelque chose qui a tapé dedans ça s'est fissuré sur oui c'est plus que quatre centimètres
AGENT: Ouais si ça dépasse les quatre centimètres ça nécessite un remplacement madame du pare-brise euh vous avez l'immatriculation du véhicule mais pour la prise en charge par l'assurance c'est nous qui le faisons directement d'accord pour que vous ayez en amont aussi la carte verte avec vous et la carte grise pour bien sélectionner le bon véhicule avec le bon vitrage
AGENT: Oui d'accord vous voulez la carte grise c'est ça mais là c'est comment

À partir de cette conversation, génère le prochain tour de parole de l'agent.
La réponse doit être courte et claire, tout en conservant le ton oral d'une conversation téléphonique. Le format attendu est : AGENT : Réponse"
AGENT:

Here is an excerpt from a phone call between an auto repair agent and a customer:
CLIENT: Yes, hello, I'd like to know if it's possible to repair a crack in the windshield of my car.
AGENT: Yes, it's a crack that exceeds four centimetres, ma'am, or is it smaller?
CLIENT: Oh yes, something hit it and it cracked over yes, it's more than four centimetres.
AGENT: Yeah, if it's more than four centimetres, you'll need to replace the windshield, madam. You'll have the vehicle registration, but we'll take care of the insurance directly, so you'll need to have the green card and the registration document with you to select the right vehicle with the right glazing.
AGENT: Yes, you want the registration document, but that's how it is.

From this conversation, generate the agent's next turn. The answer should be short and clear, while maintaining the oral tone of a telephone conversation. The expected format is: AGENT: Response".
AGENT:

State-of-The-Art Models

Pre-trained transformers with instruction-tuning

- **Llama 3 -** instruction-tuned language model with 70 billion parameters developed by (Meta, 2024). Optimized for dialogue use cases.
- **Not open: chatGPT Based** on a variant of GPT-3 named GPT-3.5, now on GPT-4 trained on conversational content (?)

Requires Instruction corpora :

Prompt/answer pairs

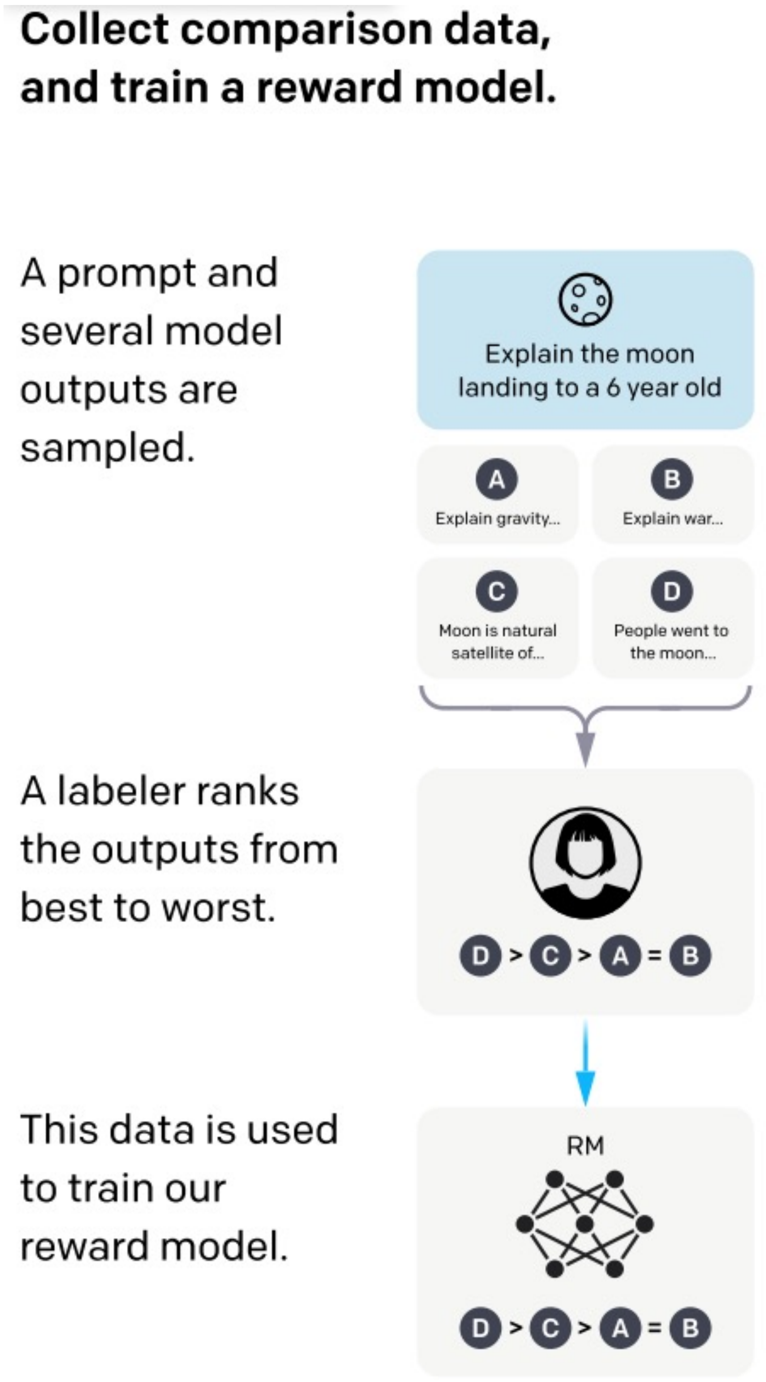
Improve the outputs

For example: focus on the safety of the answer in order to avoid inappropriate answers and social norm violations by using annotated data and external knowledge sources

- Using filtering and reranking the generated output in Google Lamda (Thoppilan et al., 2022).
- Using **reinforcement learning** by Human Feedback in chatGPT (see previous lectures)

Filtering and reranking approach from (Thoppilan et al., 2022)

- Predict the SSI (Sensibleness, Specificity, Interestingness) and Safety scores of the generated candidate responses
  - *"What's up? RESPONSE not much. SENSIBLE 1"*
  - *"What's up? RESPONSE not much. INTERESTING 0"*
  - *"What's up? RESPONSE not much. UNSAFE 0"*
- Filter out unsafe candidate responses (given a threshold)
- Reranking the filtered responses according to the following score
  - 3 * P(sensible) + P(specific) + P(interesting)
- Keep the top ranked candidate



Reinforcement learning from human feedback

STEP 2a: REWARD MODEL

Collect comparison data:

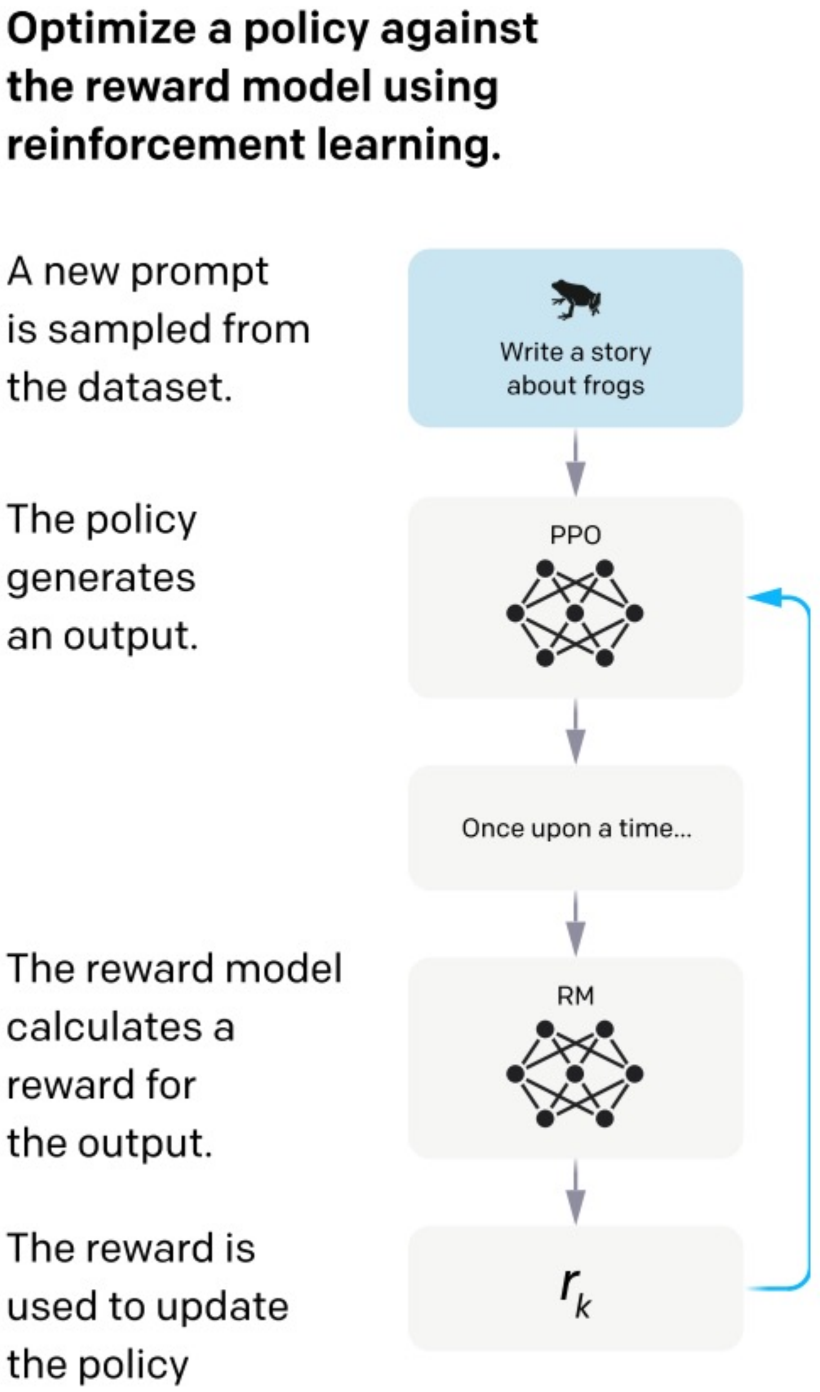
Given the SFT model of step 1, generate the n^{th} best answers (with n from 4 to 9) to a same prompt

Ask a labeler to rank the outputs

Train a **reward model** (RM) on this data:

a model able to provide a score to each (prompt,answer) pair that could reproduce the labeller ranking

Note: the reward model is trained to reproduce the preference of labellers



Reinforcement learning from human feedback

STEP 2b: Proximal Policy Optimization (PPO)
Policy : a strategy that the machine learns to use to achieve its goal
Policy optimisation: updating the model’s policy as each response is generated according to
***A Kullback-Leiber penalty** to reduce the distance of the response from the SFT model output of step 1
***The reward value computed in step 2a.**

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  - focus on modular approaches
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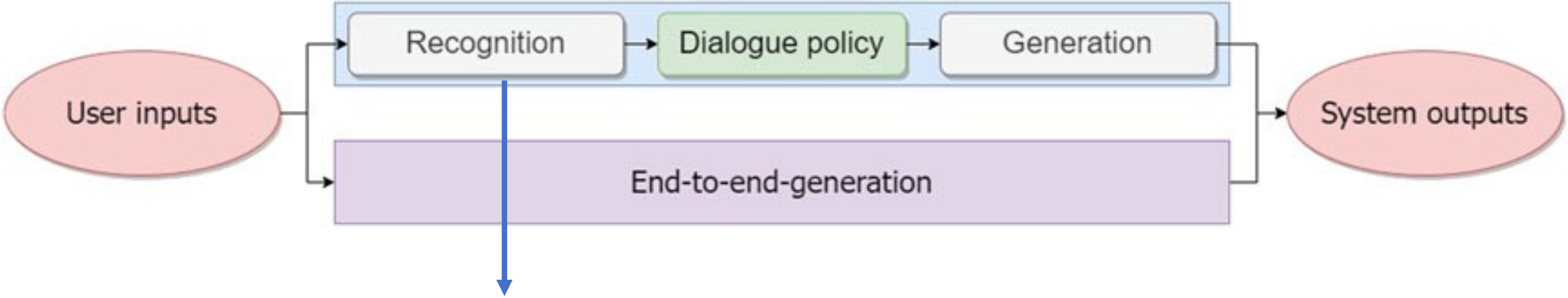
Recognition module



- Natural Language Understanding Module
- Assigning the user input to **labels** (what the system wants to detect) :
  - Speech acts/Dialogue acts (ex: **asking, inform)**
  - Intents (ex: ask for tomorrow's weather)
  - Emotions, social stances (ex: **frustration**)

Example: "can I have a train ticket to go to Paris?"

surface act = question

profound act = request

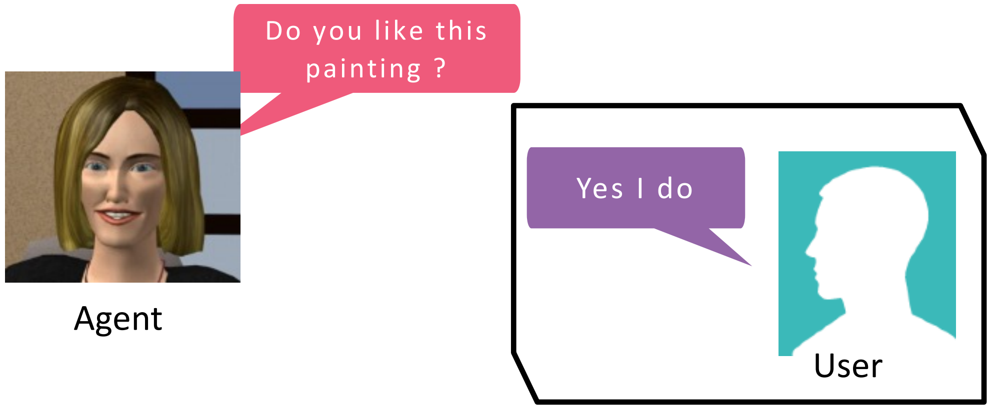
Simple example: user utterance analysis using rule-based approaches

- define rules capturing regularities/patterns that, on the basis of strings of words, reveal correlations with labels (Neviarouskaya et al., 2007; Taboada et al., 2011; Langlet and Clavel, 2015).
- rely on research in linguistics on the content of language
  - its semantics and syntax,
  - and how they are deployed in expressing intents, speech acts, stances and sentiments
    - For example, a speaker might use negation, intensifiers, conditional tense, or metaphors to express sentiments such as negativity.

Recognition module

- May integrate the visible/audible behaviors (Schuller et al., 2002; Clavel et al., 2008; Zadeh et al., 2017), such as
    - prosody,
    - facial expressions,
    - hand gestures and
    - eye gaze shifts.

- May integrate **dialogue history**

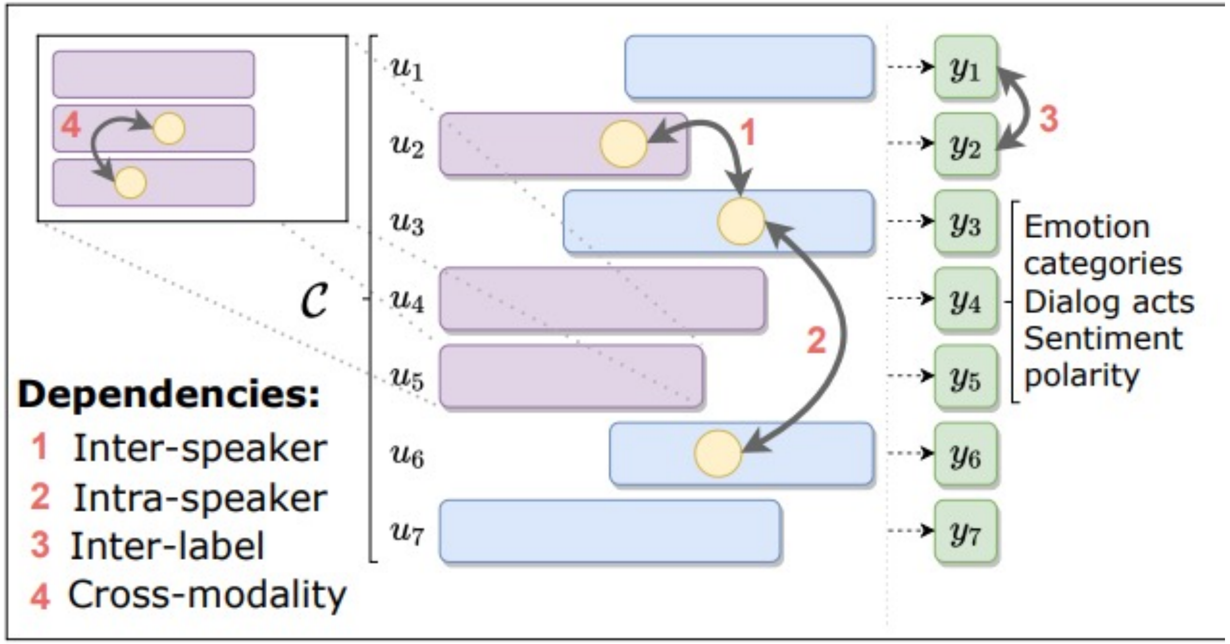


Approaches integrating dialogue history

- Modeling the dialogue context: storing what has been said, previously
  - In the initial approaches (before LLM)
    - try to interpret dialog history
    - contextual interpretation results are formalized and stored in a dedicated structure (dialog state)
    - highlighting in this history a structure describing the dialogue’s progression
      - for example, the path followed to solve the task
  - In the recent approaches
    - Build a representation of the context using neural encoders

Context modelling using neural approaches

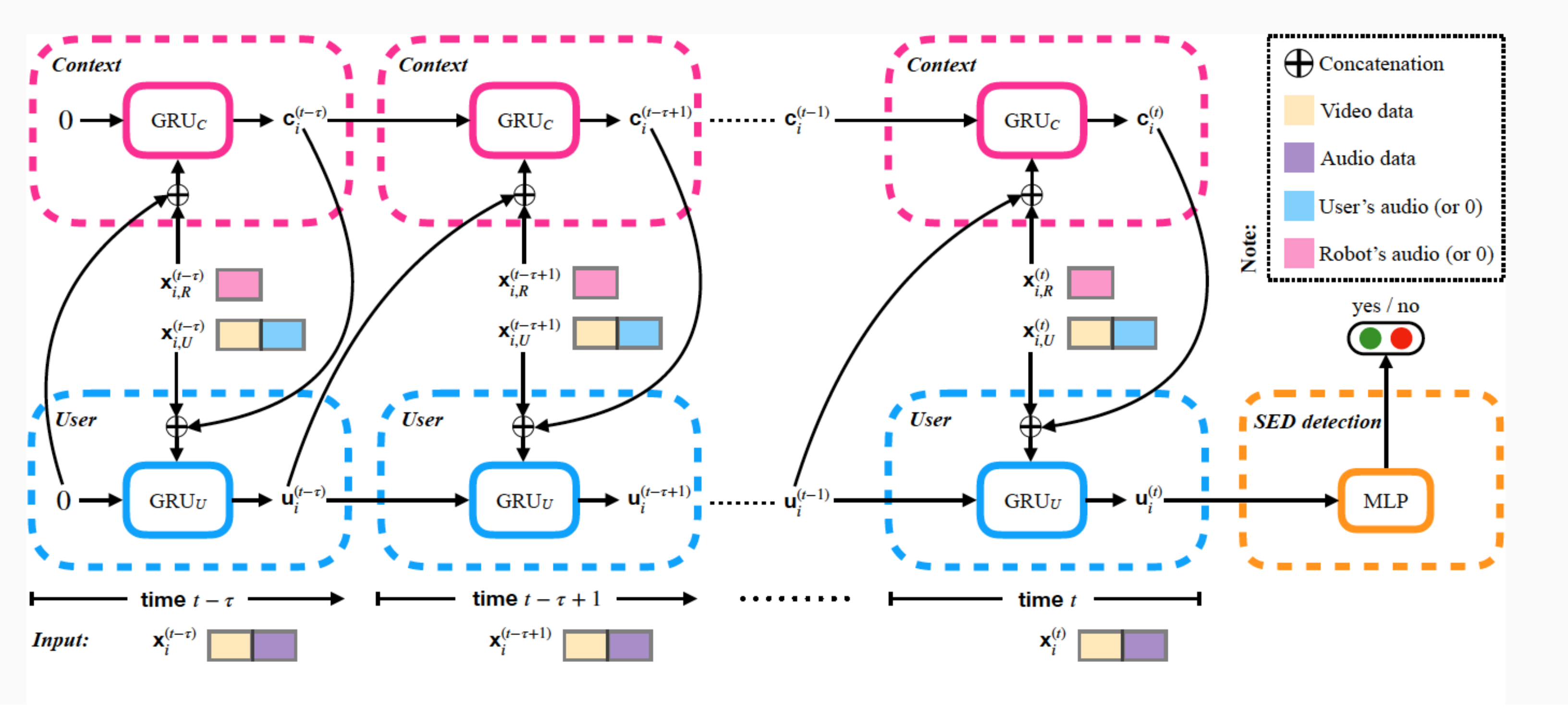
Model the conversation dynamics in the context



Context modelling using neural approaches

Atamna, A. and Clavel, C.,. HRI-RNN: A User-Robot Dynamics-Oriented RNN for Engagement Decrease Detection. In INTERSPEECH 2020.

Task : user engagement automatic analysis in a human-robot interaction
Robot data = contextual information to help assess the user's state
Architecture: HRI-RNN (Human-Robot-Interaction-Recurrent Neural Networks)

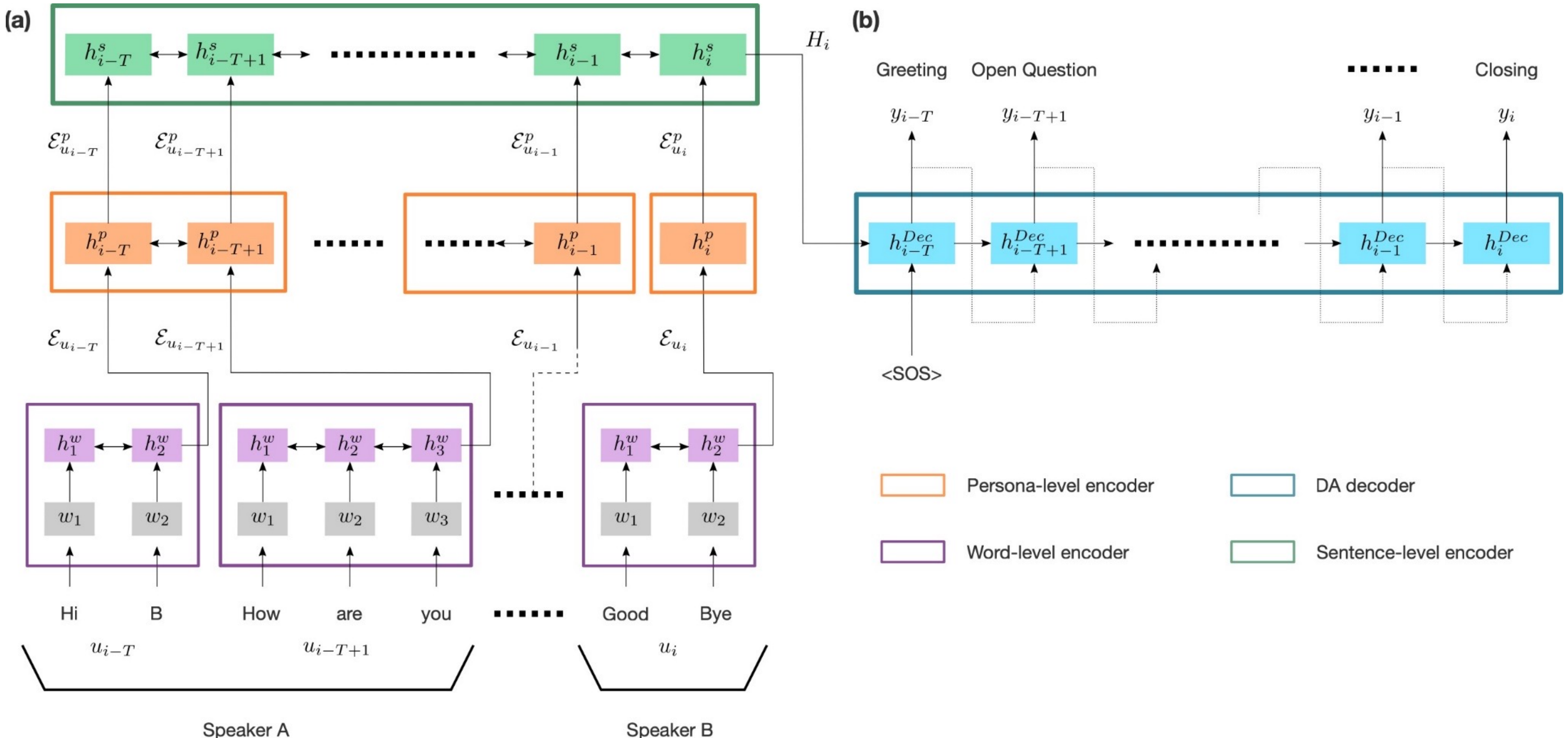


Use of a variant of RNN: GRU: Gated Recurrent Units
Build two GRUs: GRU_U to model user state and GRU_C to model context (robot data and the user's previous states + make the two GRUs interact)

Context modelling using neural approaches

Colombo et al. Guiding attention in **Sequence-to-sequence models** for Dialogue Act prediction, AAAI 2020

Integrating the conversational aspects and contextual dependencies between the utterances, the words within the utterances and the labels



Approaches integrating dialogue history

- Ideally, modeling the common ground
  - **communal common ground**, where the shared knowledge goes beyond the interaction between the speaker and the hearer
  - **personal common ground** in which the knowledge is only good for the speaker and the hearer
  - modeling the **grounding process**:
    - In the initial approaches
      - 1.automatically update the common ground at each utterance, 2. only when the interlocutor shows that they have understood (e.g. “Okay”) (discourse contributions model),
    - In the recent approaches
      - Let the network try to model implicitly this information from many observations of data

An example of grounding process: repair in human-human interactions

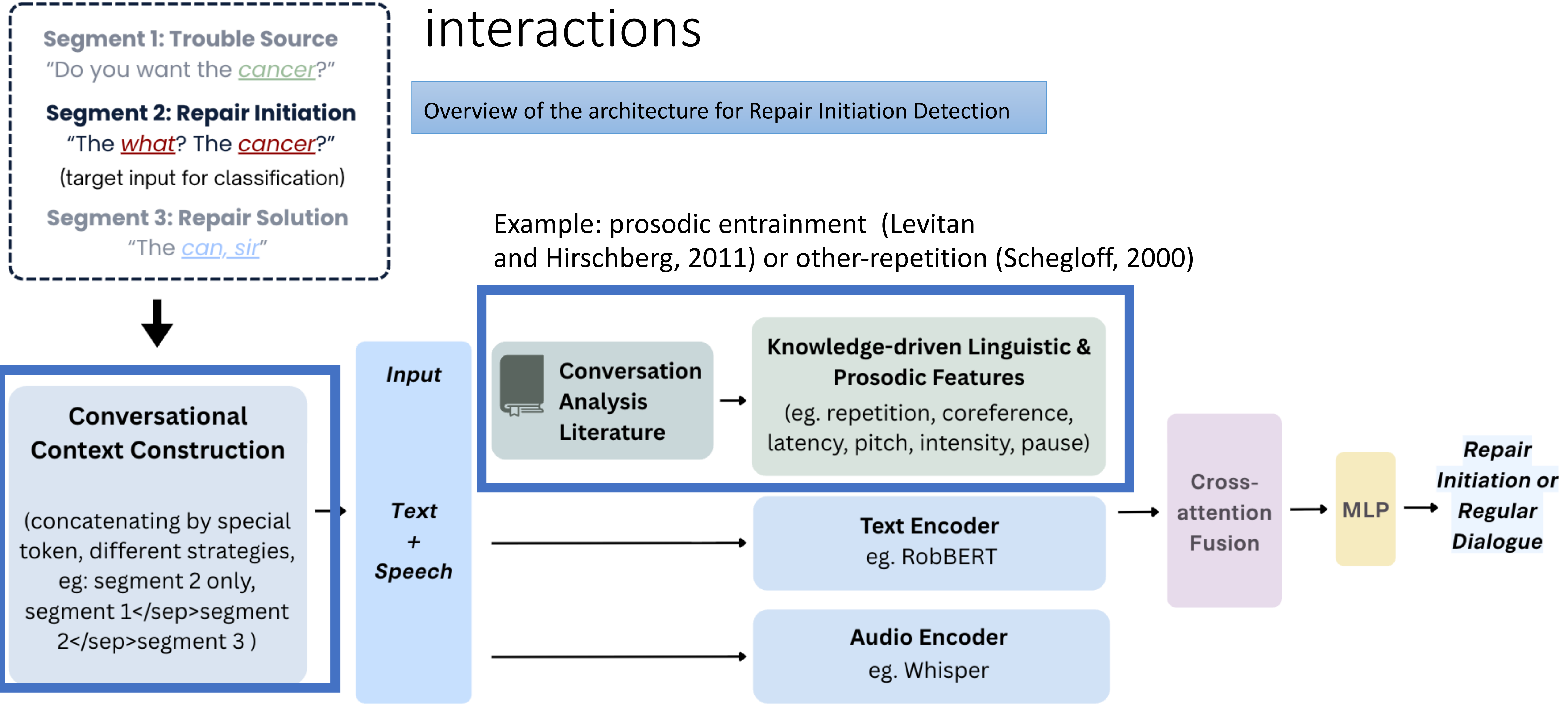
Objective: build a computational multimodal model for detecting other-initiated repair

Ngo, H.A., Rollet, N., Pelachaud, C., Clavel, C. « Mm, Wat? » Detecting Other-intiated Repair Requests in Dialogue. EMNLP 2025

Models to detect repair in human-human interactions

Overview of the architecture for Repair Initiation Detection

Example: prosodic entrainment (Levitan and Hirschberg, 2011) or other-repetition (Schegloff, 2000)



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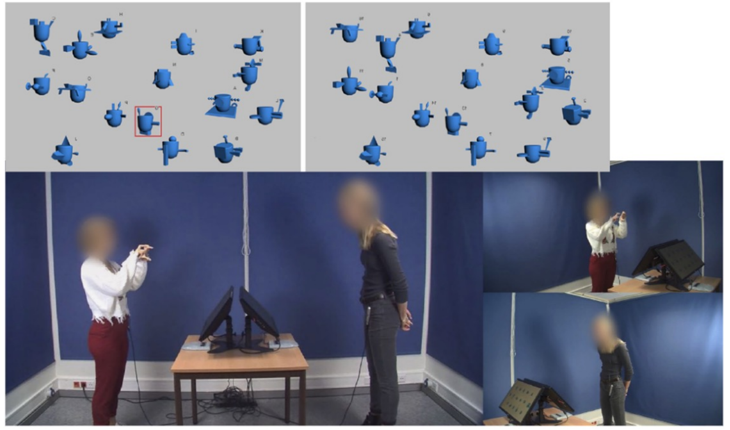
Models to detect repair in human-human interactions



Data: CABB-S corpus, in Dutch, object matching task (left: director vs right: matcher), ~7 hours recordings

Model	Modal & Features	Precision	Recall	F1-score
Text_{Emb}	U & T	72.0 ± 4.0	87.6 ± 7.5	78.9 ± 4.7
$\text{Audio}_{\text{Emb}}$	U & A	72.6 ± 9.7	76.3 ± 13.1	70.6 ± 8.1
$\text{Multi}_{\text{Emb}}$	M & T+A	79.1 ± 5.4	82.2 ± 3.8	82.1 ± 0.9
$\text{Text}_{\text{Ling}}$	U & L	82.2 ± 3.6	80.4 ± 6.1	80.4 ± 3.8
$\text{Audio}_{\text{Pros}}$	U & P	81.7 ± 4.2	77.4 ± 5.4	77.3 ± 2.7
$\text{Multi}_{\text{LingPros}}$	M & L+P	81.7 ± 7.6	82.2 ± 1.5	81.8 ± 3.4
$\text{Multi}_{\text{Ours}}$	M & T+A+L+P	$\mathbf{93.2 \pm 2.8}$	$\mathbf{96.1 \pm 2.6}$	$\mathbf{94.6 \pm 2.3}$

U: Unimodal, M: Multimodal, T: Text, A: Audio, P: Prosodic features, L: Linguistic features. Overall results: $\text{Multi}_{\text{Ours}} > \text{Multi}_{\text{Emb}} \sim \text{Multi}_{\text{LingPros}}$

Key takeaways:

- **multimodal models** performed better than unimodal
- **designed linguistic and prosodic features** ≈ text embeddings and audio embeddings
- Significant improvement **when combining knowledge-driven features and embeddings** (+12.5 pp)

Ngo, H.A., Rollet, N., Pelachaud, C., Clavel, C. « Mm, Wat? » Detecting Other-intiated Repair Requests in Dialogue. EMNLP 2025

Role of future context in the repair detection

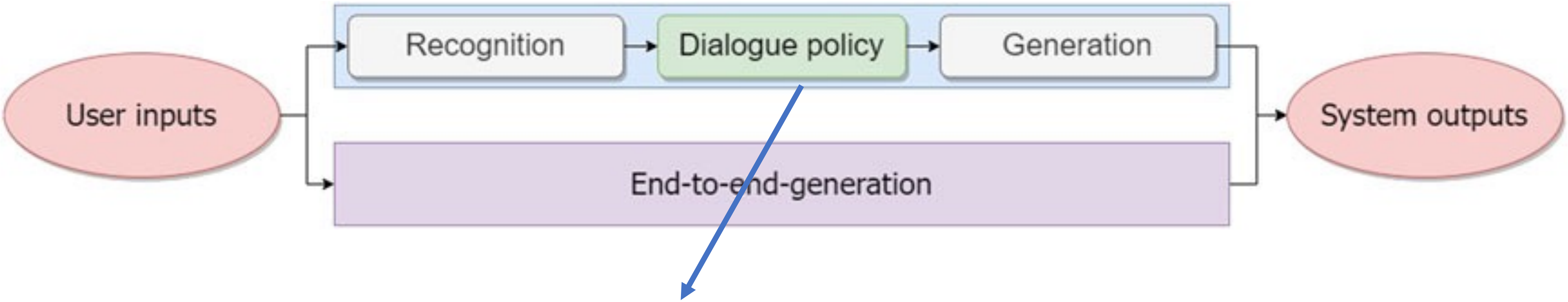
Context	Win. len	Precision	Recall	F1-score
(1) $\text{Past}_{\text{Context}}$	2	86.0 ± 3.0	78.4 ± 5.4	82.0 ± 4.1
(1) $\text{Past}_{\text{Context}}$	max	86.6 ± 5.2	81.0 ± 6.1	83.5 ± 4.3
(2) $\text{Current}_{\text{Context}}$	-	84.6 ± 3.8	82.9 ± 6.0	83.6 ± 4.4
(3) $\text{Future}_{\text{Context}}$	max	84.00 ± 1.53	78.20 ± 5.78	80.18 ± 2.52
(4) $\text{Multi}_{\text{Ours}}$	2	$\mathbf{93.2 \pm 2.8}$	$\mathbf{96.1 \pm 2.6}$	$\mathbf{94.6 \pm 2.3}$
(4) $\text{Multi}_{\text{Ours}}$	max	87.7 ± 3.5	89.1 ± 5.3	88.3 ± 3.7

Table 2: Performance comparison across different micro context configurations

Outline

- Background
- Approaches:
  - focus on end-to-end approaches
  - focus on modular approaches – dialogue management
- Data and evaluation
- Applications and ethical issues

Dialogue policy/management module



Modular/Dialogue policy module

Selecting the type of answer given the previous dialogue understanding outputs;

Examples of types of answer:

- An *Intent* , ex : a suggestion of a particular train
- A suitable *socio-emotional strategy*

Figure ©Chenwei Wan,internship student 2024

Example of socio-emotional strategies

From Vanel, L., Yacoubi, A., Clavel, C. (2023) A Survey of Socio-Emotional Strategies for Generation-Based Conversational Agents, ICAART

Table 1: List of dialogue strategies.

Dialogue Strategy	**Definition**
Agreeing	Showing agreement with the interlocutor *(Rashkin et al., 2018; Welivita et al., 2021)*
Informing	Giving relevant information *(Liu et al., 2021)*
Questioning	Asking a direct question *(Rashkin et al., 2018; Welivita et al., 2021; Liu et al., 2021)*
Back-Channeling	Interjecting to signal attention *(Hardy et al., 2021)*
Consoling	Cheering someone up if they show negative sentiment *(Rashkin et al., 2018; Welivita et al., 2021)*
Encouraging	Showing enthusiasm and support *(Rashkin et al., 2018; Welivita et al., 2021; Liu et al., 2021)*
Providing Suggestions	Giving advice on how to react to a situation *(Rashkin et al., 2018; Welivita et al., 2021; Liu et al., 2021)*
Restating	Reformulating the interlocutor's point *(Liu et al., 2021)*
Self-Disclosing	Revealing information about oneself *(Liu et al., 2021; Soni et al., 2021; Hardy et al., 2021)*
Sympathising	Reflecting the interlocutor's feelings *(Rashkin et al., 2018; Welivita et al., 2021; Liu et al., 2021)*

Dialogue management

Different options to select the answer type:

- only look at a single prior turn by the user (and its analysis by the recognition module)
- use external information sources (knowledge bases)
  - This is the case in some question-answering systems
- rely on some kind of longer history
  - ex : 5 preceding speaker turns such as in (Albumiti et al., 2023)
  - (and their analysis by the recognition module)
- Define a **planning structure** that establishes how to move from one dialogue task to another in order to achieve the user's goals
  - Ex: hierarchical task network (Rich and Sidner, 1997).

[How About Kind of Generating Hedges using End-to-End Neural Models?]
(Abulimiti, Clavel and Cassell, ACL 2023)

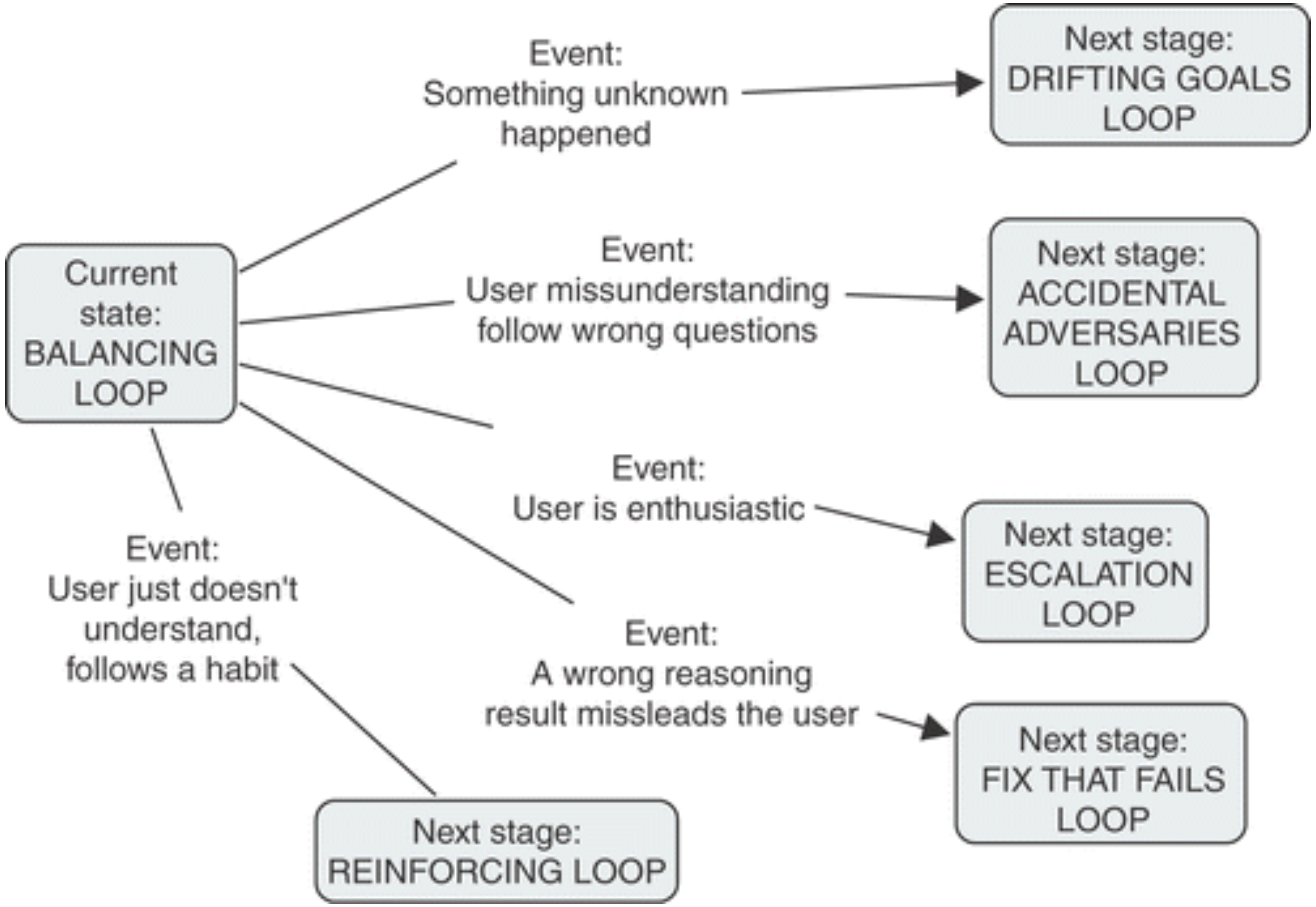
Methods

- Initial approaches
  - Reasoning models
  - Statistical approaches with Reinforcement learning
- More recent approaches
  - Supervised machine learning
  - llm with prompting

Reasoning models

Example of studies:

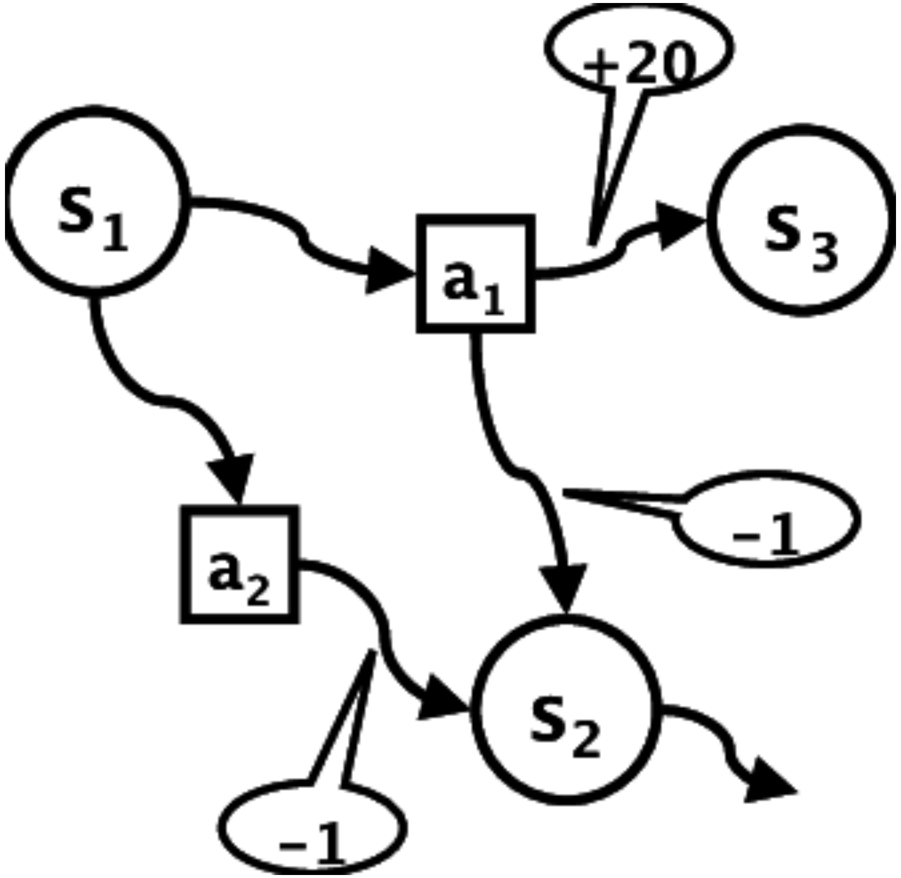
- Use of ontologies for Conversational Case-based reasoning COBBER [Gaumez-Gauchia et al., 2006]



Learning a policy using a reward : MDP Markov decision processes

- A dialog is represented as a sequence of
  - States
    - "I want a restaurant in Paris."
    - -> State :

      intent = book_restaurant
      city = Paris
      time = ?
      people = ?
  - And actions :
    - Ex: Ask_time, search_restaurant, book_table
- At each time t, the Dialog Manager
  - is in state st,
  - takes action at,
  - transitions into state st+1
  - and receives a reward rt+1.
    - for example action « book_table » that finalize the task can get +20, and « ask_time » -1 to penalize long dialog



From Schatzmann et al.. (2006). A survey of statistical user simulation techniques for reinforcement-learning of dialogue management strategies.

Learning a policy using a reward : MDP Markov decision processes

- User simulation to learn a dialogue strategy π(s) → a : can be viewed as a deterministic mapping from states to actions
- Simulate a user with
  - a goal
    - Ex:
      book_restaurant
      city = Paris
      time = 19:00
      people = 4
  - and an agenda (list of response to the agent)
    - inform(city)
      inform(time)
      inform(people)
      confirm_booking
- The simulator chooses the next response among the remaining responses according to the system action

From Schatzmann et al.. (2006). A survey of statistical user simulation techniques for reinforcement-learning of dialogue management strategies.

Statistical approaches including a reward

(POMDPs) - Partially Observable Markov Decision Process,

- A generalization of MDP
- The state s_t is not directly observable reflecting the uncertainty in the interpretation of user utterances (for example because of ASR errors)
- Instead the model considers a noisy observation o_t of the user input with probability p(ot/st)
- As the dialog progresses, a reward is assigned at each step designed to mirror the desired characteristics of the dialog system.
- The dialog model M and the policy model P can then be optimized by maximizing the expected accumulated sum of these rewards, either …
  - **online** through interaction with users or
  - **offline** from a corpus of dialogs collected within a similar domain.

From Young et al., "POMDP-Based Statistical Spoken Dialog Systems: A Review,"

More recent approaches: Supervised machine learning for next strategy prediction

Task formalization: Learn to predict next label (dialog policy of next utterance)

- Reminder
  - Training data: set of **labelled** dialogues $\mathrm{D} = \{\mathrm{d}_1, \mathrm{d}_2, \mathrm{d}_3, \dots \mathrm{d}_\mathrm{n}\}$
  - Each dialogue $\mathrm{d} = \{\mathrm{u}_1, \mathrm{u}_2, \mathrm{u}_3 \dots \mathrm{u}_\mathrm{m}\}$ consists of m turns
    - u_i is a vector $(\mathrm{x}_1, \mathrm{x}_2, .., \mathrm{x}_\mathrm{N})$ (N is the size of the representation space)
    - $h_i = \{u_{max(1,i-\omega)}, u_{i-\omega+1}, \dots u_i\}$ dialogue history (of fixed window size (ω))
- l_i the label of a particular turn u_i.

Training objective:

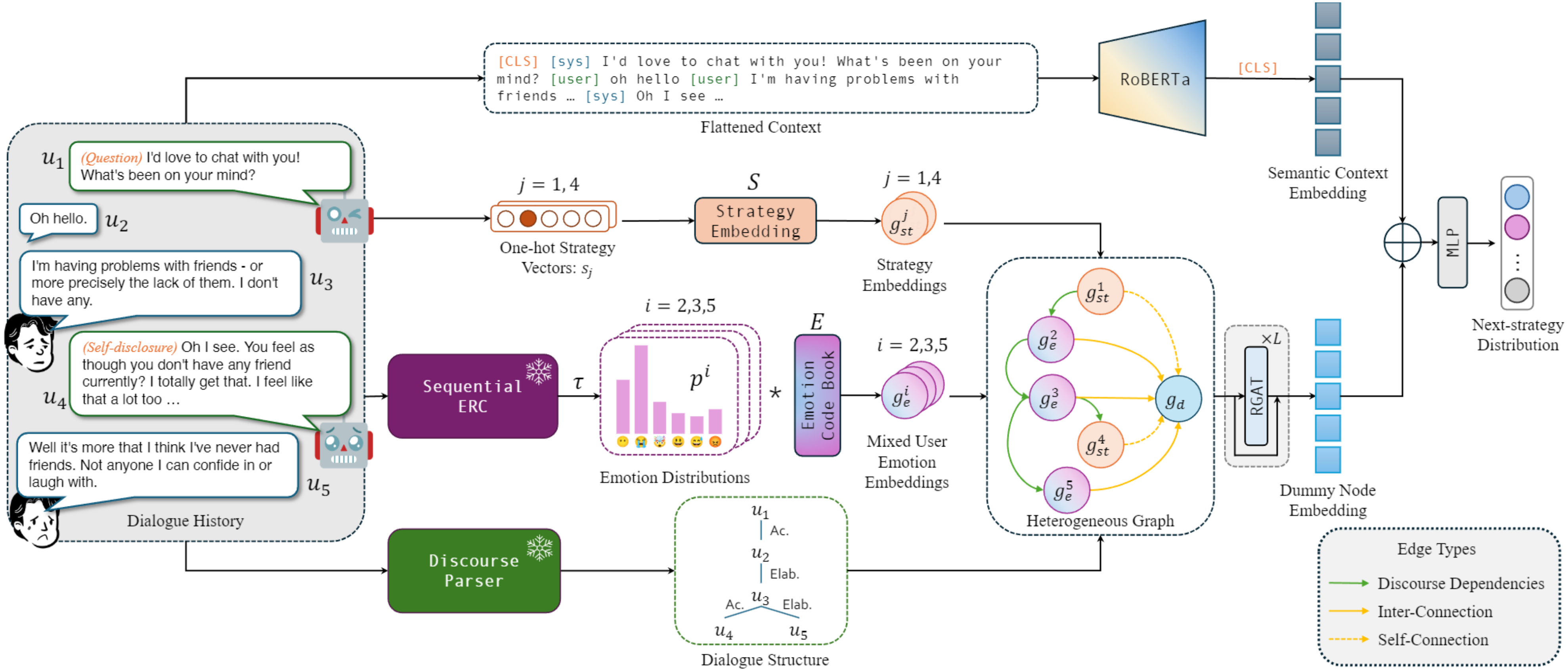
- find a model M, capable of predicting the next label l'_{i+1} based on the dialogue history h_i

SOTA Approaches : Transformers used as a classifier with or without graph structure

(Wan, Labeau and Clavel, NAACL 2025)

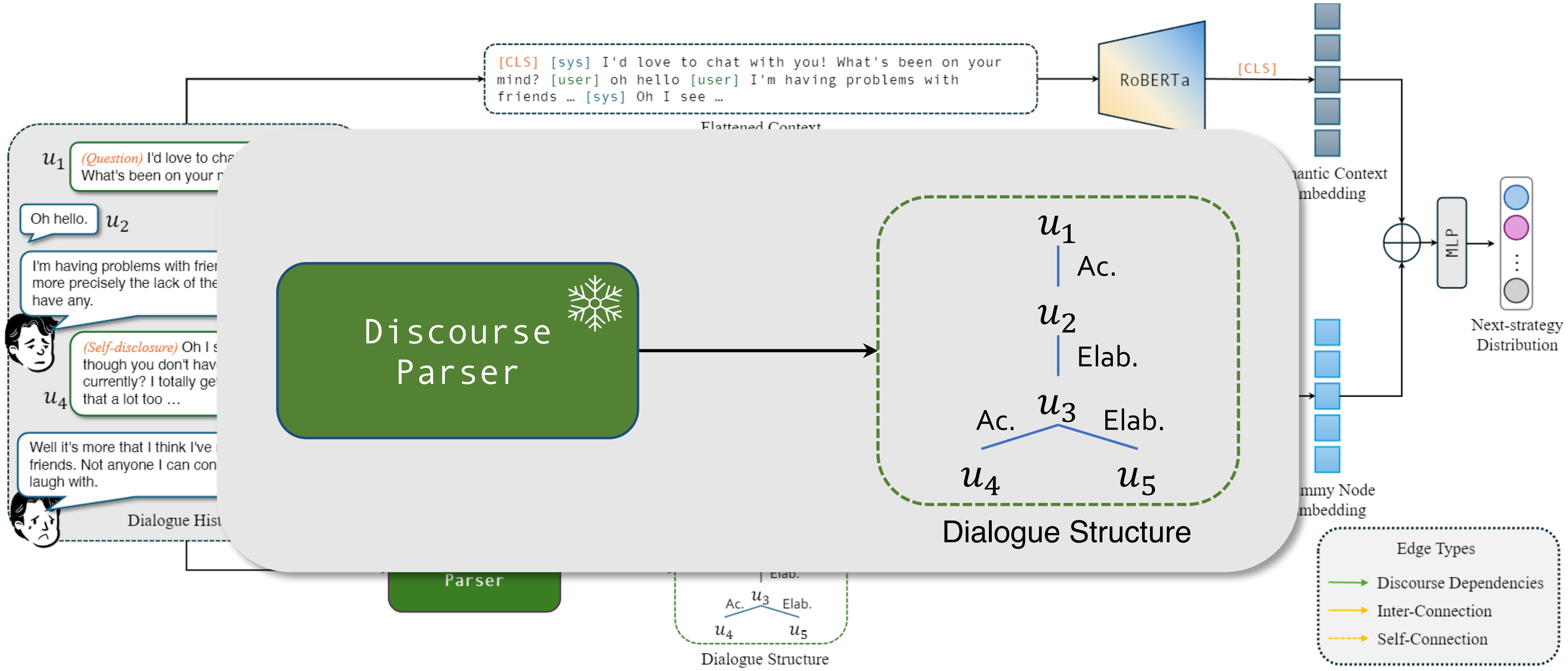
Graph neural networks for modelling the conversational dynamics of past history

- EmodynamiX model



Graph neural networks for modelling the conversational dynamics of past history

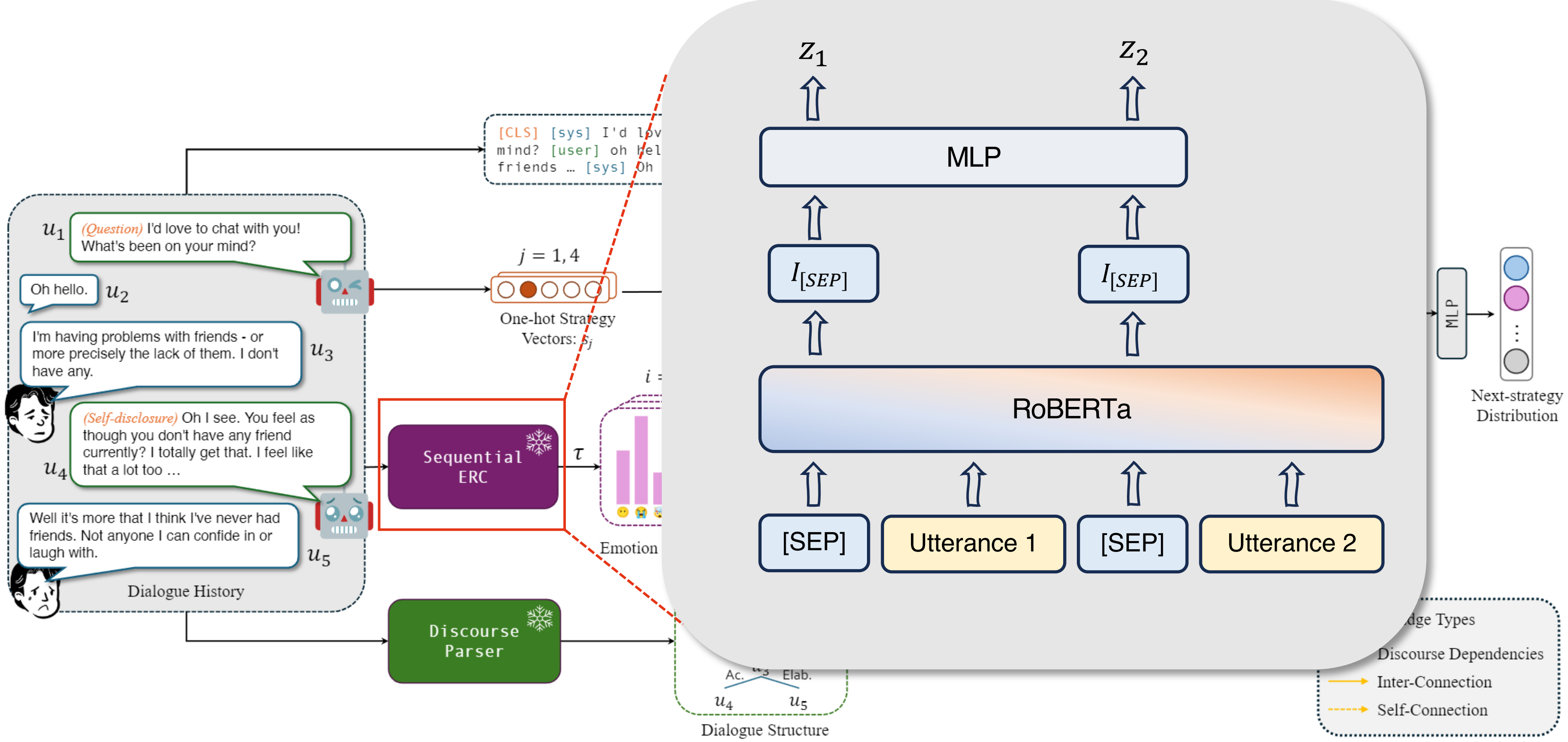
Off-the-shelf <u>discourse parser</u> to identify relationships between past utterances



Example dependency types: *Acknowledgment, Elaboration, Correction …*

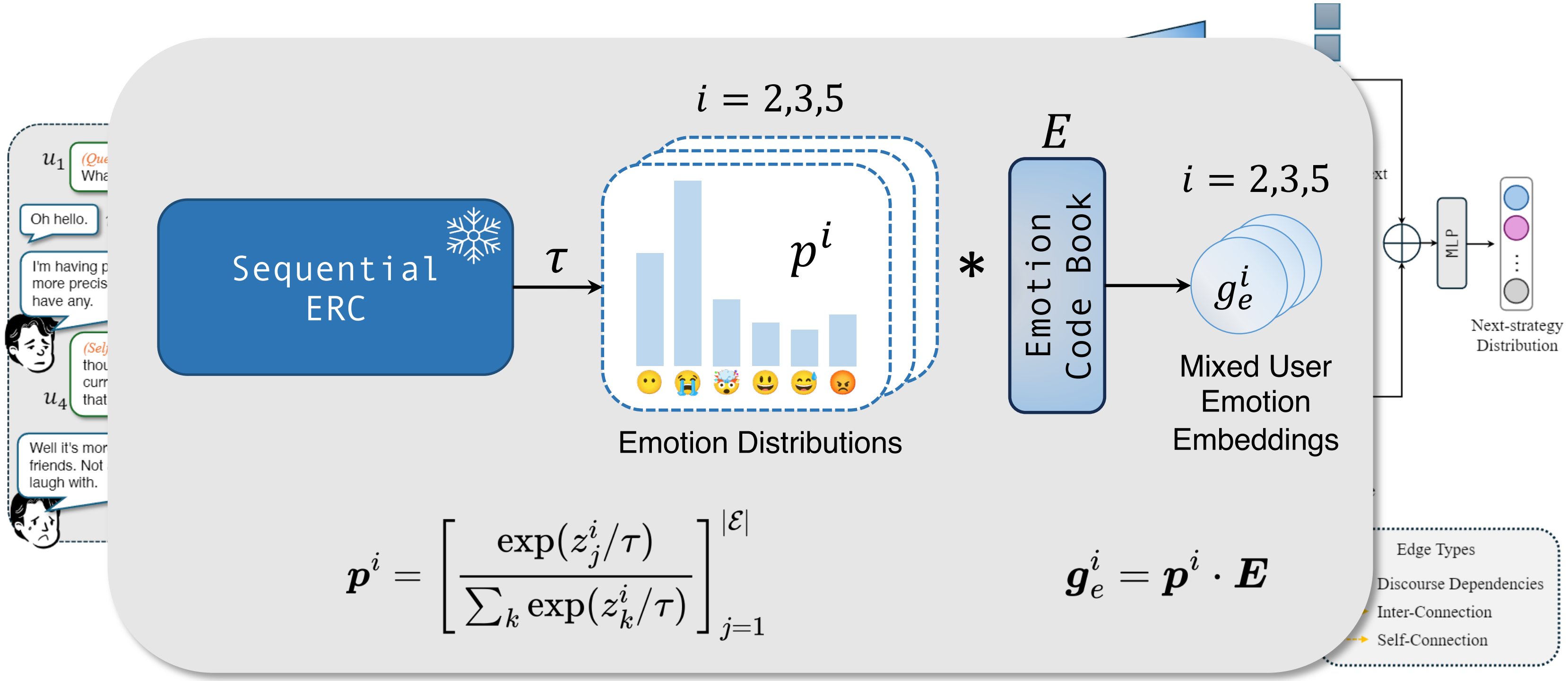
Graph neural networks for modelling the conversational dynamics of past history

Pre-trained emotion recognition model



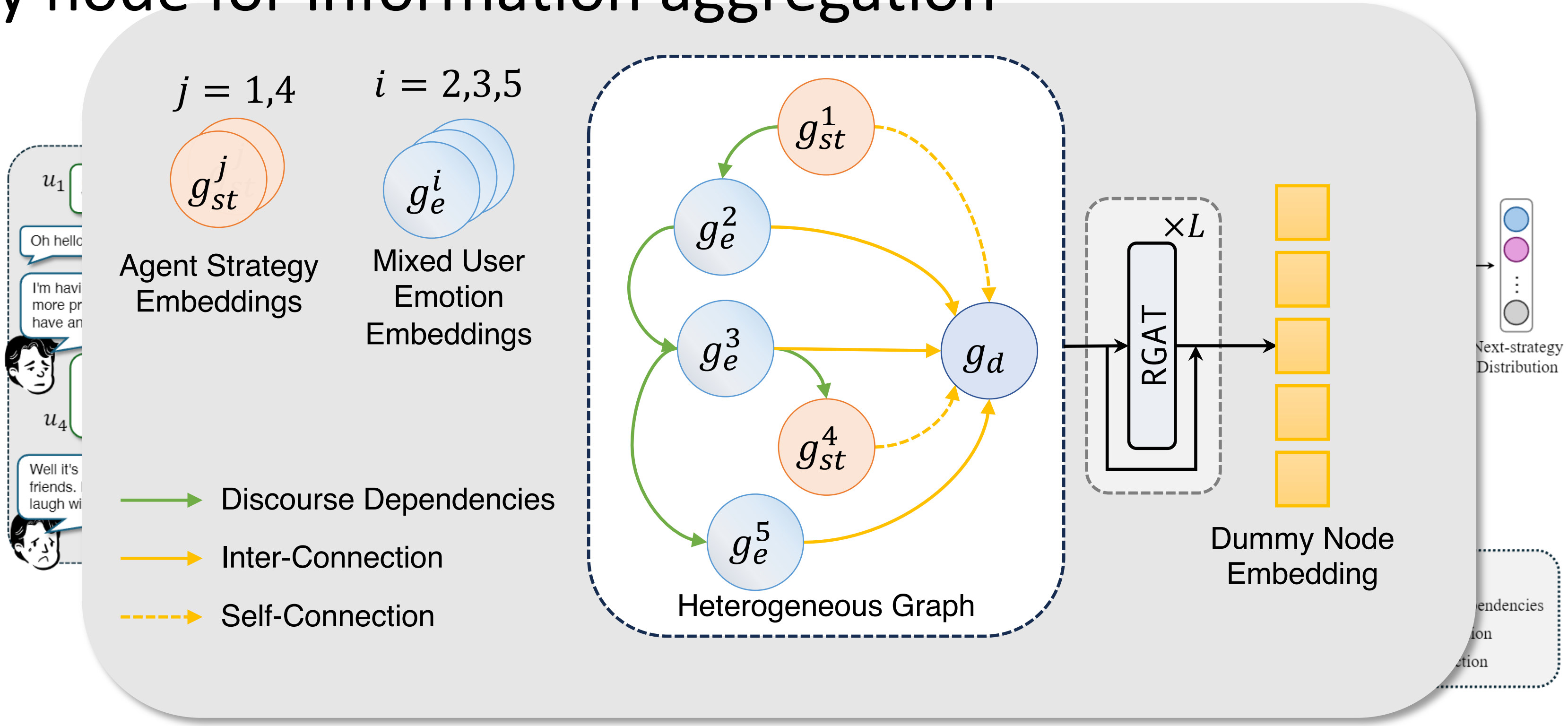
Graph neural networks for modelling the conversational dynamics

Off-the-shelf emotion recognition model in order to understand the emotional content of the past utterances of the user.



Graph neural networks for modelling the conversational dynamics

- relational graph attention network + heterogeneous graph with dummy node for information aggregation



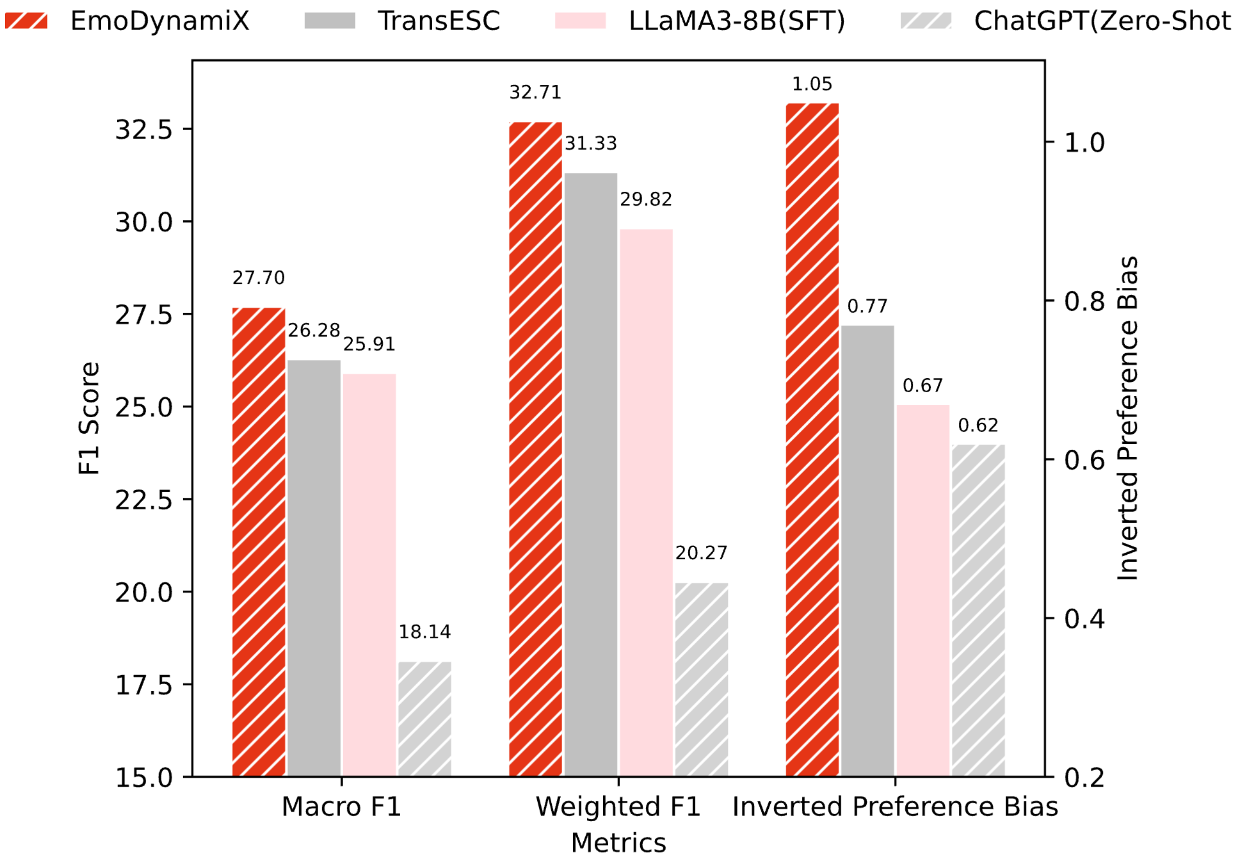
Graph neural networks for modelling the conversational dynamics

EmoDynamiX outperforms:

- **SOTA frameworks**: TransESC (Zhao et al., 2023), etc.
- **Finetuning SOTA LLMs**: LLaMA3-8B (Meta, 2024), etc.
- **Prompting SOTA LLMs**: ChatGPT (OpenAI, 2023), etc.

Across all metrics on two datasets (ESConv and AnnoMI).

Results on ESConv



Inverted Preference Bias = 1.5 – Preference Bias

LLM with prompting for action prediction

From Wan, C., Labeau, M., & Clavel, C. (2024).

Question: asking for information related to the problem to help the seeker articulate the issues that they face.
Restatement or Paraphrasing: a simple, more concise rephrasing of the seeker's statements that could help them see their situation more clearly.
Reflection of Feelings: describe the help-seeker's feelings to show the understanding of the situation and empathy.
Self-disclosure: share similar experiences or emotions that the supporter has also experienced to express your empathy.
Affirmation and Reassurance: affirm the help-seeker's ideas, motivations, and strengths to give reassurance and encouragement.
Providing Suggestions: provide suggestions about how to get over the tough and change the current situation.
Information: provide useful information to the help-seeker, for example with data, facts, opinions, resources, or by answering questions.
Others: other support strategies that do not fall into the above categories.

Prompting Template

Task Description
You are an intelligent emotional support assistant dedicated to helping people cope with stress and depression. To effectively comfort your users, you must select the appropriate dialogue strategy based on the context of the conversation and the user's emotional state. Choose from the following x types of strategies:
Strategy Descriptions
...
Example 1
Dialogue context
...
Output
...
Example 2
...
Dialogue Context
supporter: (*strategy*) ...
seeker: (*emotion: optional*)...

Task
Now select the appropriate dialogue strategy for the next utterance according to the task description and dialogue context above (one answer only, no description), output should be in this format: (*strategy*).

Outline

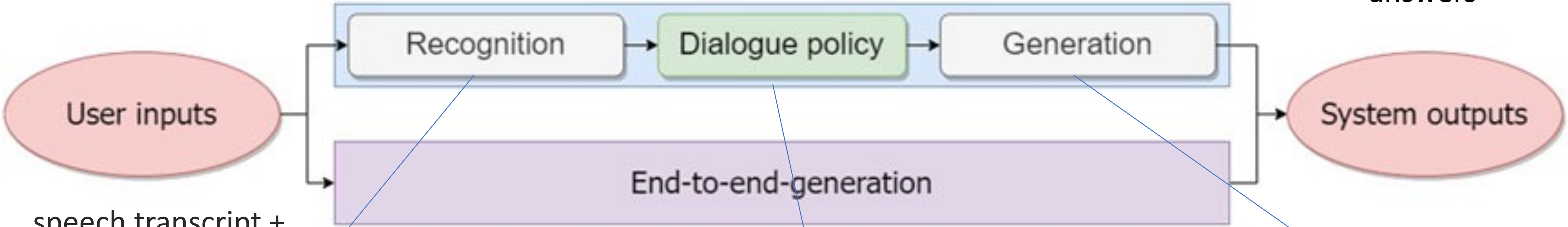
- Background
- Approaches:
  - focus on end-to-end approaches
  - focus on modular approaches – generation module
- Data and evaluation
- Applications and ethical issues

How does it work?

Clavel, C., Labeau, M., & Cassell, J. (2023). [Socio-conversational Systems: Three Challenges at the Crossroads of Fields.]() Frontiers in Robotics and AI

- **Modular** vs. end-to-end approaches

Socially-relevant answers



speech transcript + visible/audible behaviors

Assigning the user input to labels
(ex: an intent - *book a train*
or an emotion - *frustration*)

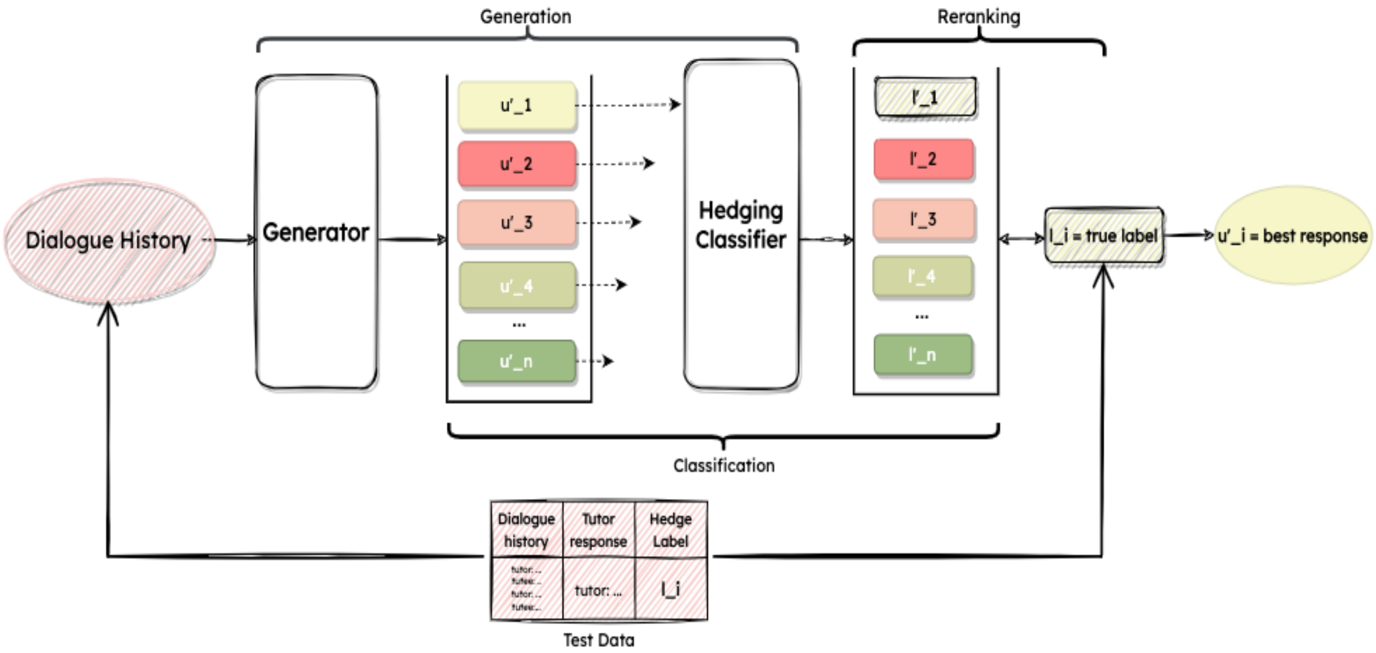
Selecting the type of answer
(ex. an intent
- a suggestion of a particular train
or a suitable socio-emotional strategy
– empathy)

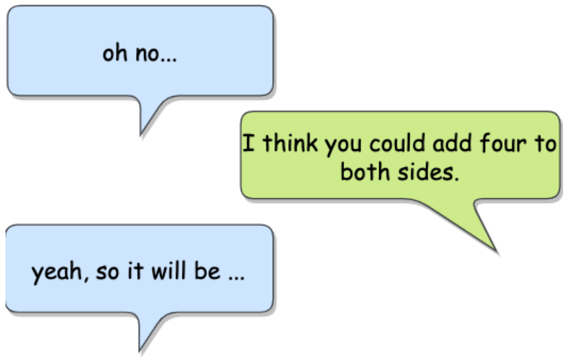
Generating the text and visible/audible behaviors corresponding to the selected dialogue policy

Generation module

- The generation of natural language in these modular systems is accomplished with
  - Templates
  - or language models
- When templates:
  - Ex: Eliza or TRAINS
- When language models:
  - Filtering the answer that is the closest to the desired dialogue policy
  - Or Conditional generation

Example for filtering the answers





[How About Kind of Generating Hedges using End-to-End Neural Models?]
(Abulimiti, Clavel and Cassell, ACL 2023)

Design of an architecture stacking:
- a generator (ex: DialogGPT) using dialog history
- a classifier for **filtering** the outputs forcing to generate the relevant strategy

Example of basic prompt for conditional generation in dialogue

ça serait j'ai fissuré le pare-brise de ma voiture donc j'aimerais savoir si c'est possible de pour réparer
AGENT: Oui c'est une fissure qui dépasse les quatre centimètres madame ou c'est inférieur ?
CLIENT: Ah oui il y a quelque chose qui a tapé dedans ça s'est fissuré sur oui c'est plus que quatre centimètres
AGENT: Ouais si ça dépasse les quatre centimètres ça nécessite un remplacement madame du pare-brise euh vous avez l'immatriculation du véhicule mais pour la prise en charge par l'assurance c'est nous qui le faisons directement d'accord pour que vous ayez en amont aussi la carte verte avec vous et la carte grise pour bien sélectionner le bon véhicule avec le bon vitrage
AGENT: Oui d'accord vous voulez la carte grise c'est ça mais là c'est comment

|

À partir de l'extrait de dialogue et du plan de réponse suivant, génère la réponse de l'agent:
1. Action to take 1
2. Action to take 2
La réponse doit être courte et claire, tout en conservant le ton oral d'une conversation téléphonique. Le format attendu est : AGENT : Réponse
AGENT:

know if it's possible to repair a crack in the windshield of my car.
AGENT: Yes, it's a crack that exceeds four centimetres, ma'am, or is it smaller?
CLIENT: Oh yes, something hit it and it cracked over yes, it's more than four centimetres.
AGENT: Yeah, if it's more than four centimetres, you'll need to replace the windshield, madam. You'll have the vehicle registration, but we'll take care of the insurance directly, so you'll need to have the green card and the registration document with you to select the right vehicle with the right glazing.
AGENT: Yes, you want the registration document, but that's how it is.

|

From the dialogue extract and the following response plan, generate the agent's answer:
1. Action to take 1
2. Action to take 2

|

The answer should be short and clear, while maintaining the oral tone of a telephone conversation. The expected format is: AGENT: Response.
AGENT:

Example of advanced prompt for conditional generation in dialogue

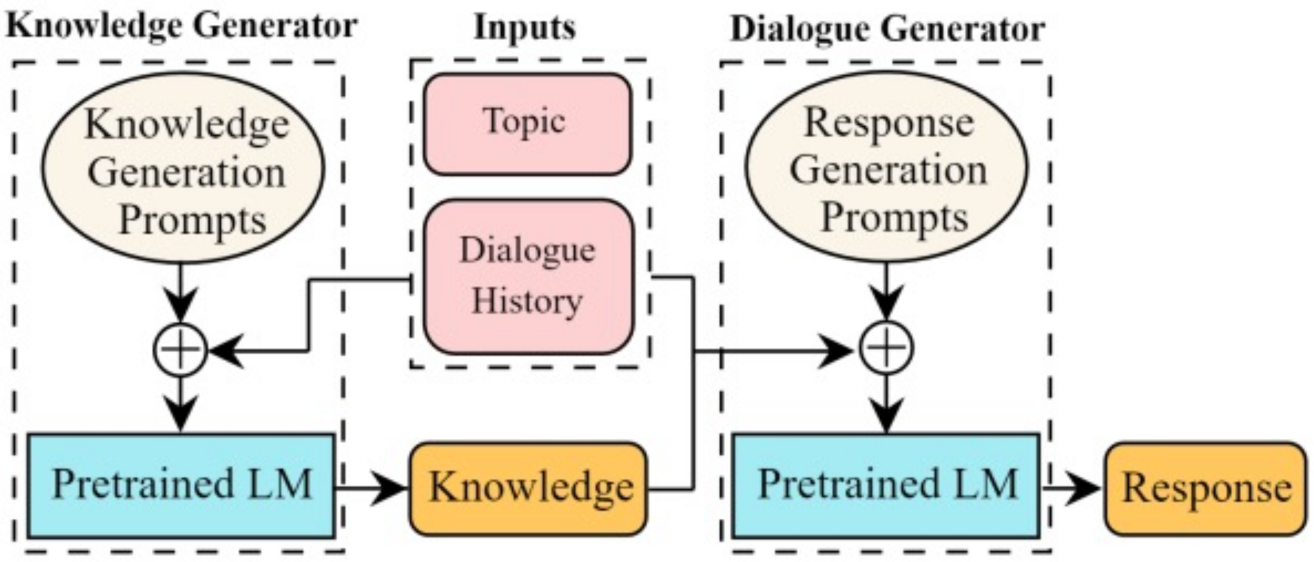


Figure 1: Our proposed framework (MSDP) for the knowledgeable dialogue generation.

Two steps:
1/Prompting LM to generate knowledge
2/Prompting LM to generate the answer conditioned to the knowledge (few-shot samples from the dataset)

[Multi-Stage Prompting for Knowledgeable Dialogue Generation](Liu et al., Findings 2022)

Example of advanced prompt for conditional generation in dialogue

1/Prompting LM to generate knowledge From Topic

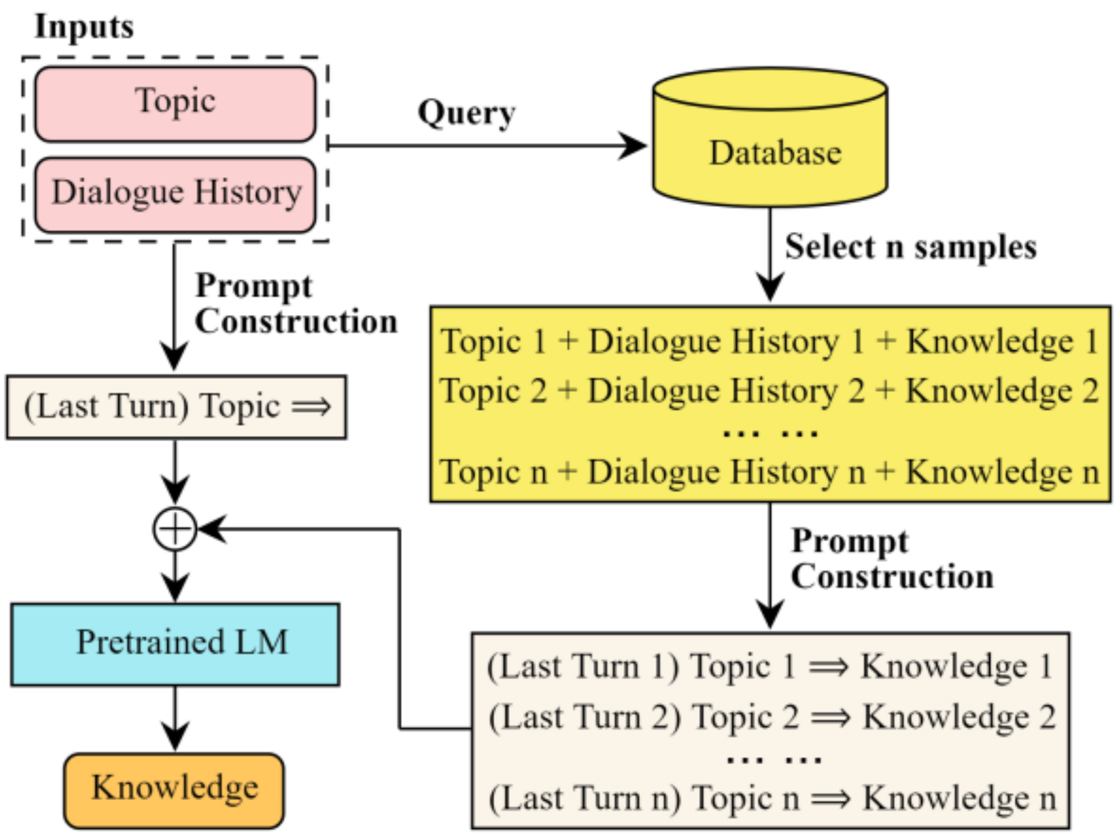


Figure 2: Prompting for the knowledge generation.

Knowledge generation : find similar samples (using knowledge grounded datasets) + construct the prompt + generate knowledge

Example of advanced prompt for conditional generation in dialogue

1/

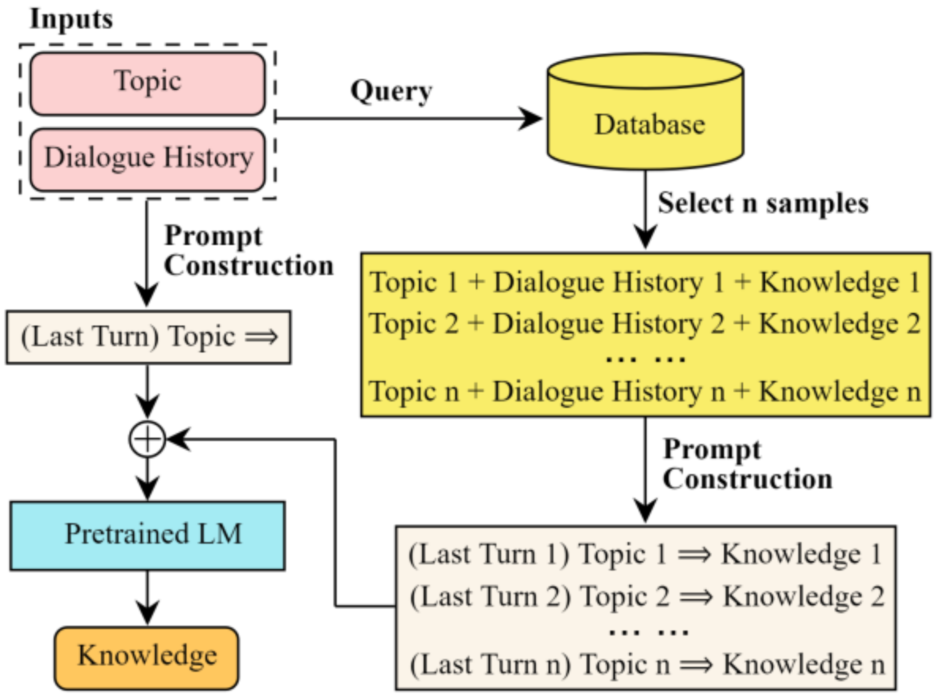


Figure 2: Prompting for the knowledge generation.

Knowledge generation : find similar samples (using knowledge grounded datasets) + construct the prompt + generate knowledge

2/

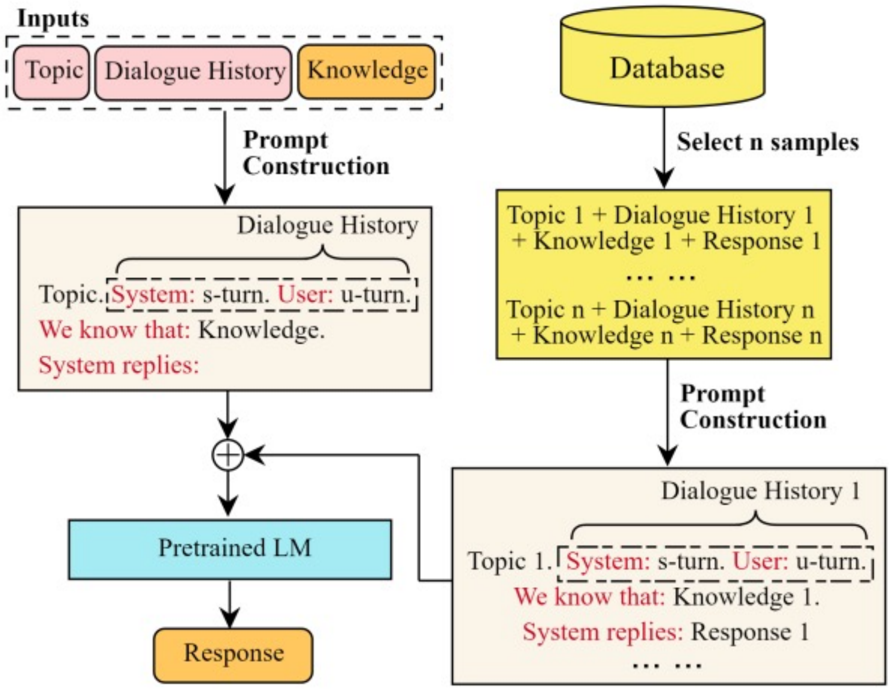


Figure 3: Prompting for the dialogue response generation. We use comprehensive words (denoted in red color) to connect the dialogue history, knowledge and response for the prompt construction.

Outline

- Background
- Approaches: focus on modular approaches
- Data and evaluation
- Applications and ethical issues

Data

- Models are trained from human-human interactions
- Examples: DailyDialog, BlendedSkillTalk, MELD, AMI corpus, OpenSubtitle, PersonaChat, WizardofWkipedia, ProSocialDialog, etc.
- Possible data augmentation by generating conversations using large language models : example SODA

[SODA: Million-scale Dialogue Distillation with Social Commonsense Contextualization](https://aclanthology.org/2023.emnlp-main.799) (Kim et al., EMNLP 2023)

DailyDialog

- **scripted dataset** (human-authored artificial conversation dataset)
- Written interactions in English
- **low level of spontaneity** : raw data crawled from various websites which serve for English learner to practice English dialog in daily life

A: I'm worried about something.
B: What's that?
A: Well, I have to drive to school for a meeting this morning, and I'm going to end up getting stuck in rush-hour traffic.
B: That's annoying, but nothing to worry about. *Just breathe deeply when you feel yourself getting upset*.
A: Ok, I'll try that.
B: Is there anything else bothering you?
A: Just one more thing. A school called me this morning to see if I could teach a few classes this weekend and I don't know what to do.
B: Do you have any other plans this weekend?
A: I'm supposed to work on a paper that'd due on Monday.
B: *Try not to take on more than you can handle.*
A: You're right. I probably should just work on my paper. Thanks!

Figure 1: An example in **DailyDialog** dataset. Some text is shortened for space. Best viewed in color.

BlendedSkillTalk

- Written interactions in English (about 5k conversations)
- **low level of spontaneity** : crowdsourced dataset where workers are instructed to try and be knowledgeable, empathetic, or give personal details about their given persona, whenever appropriate

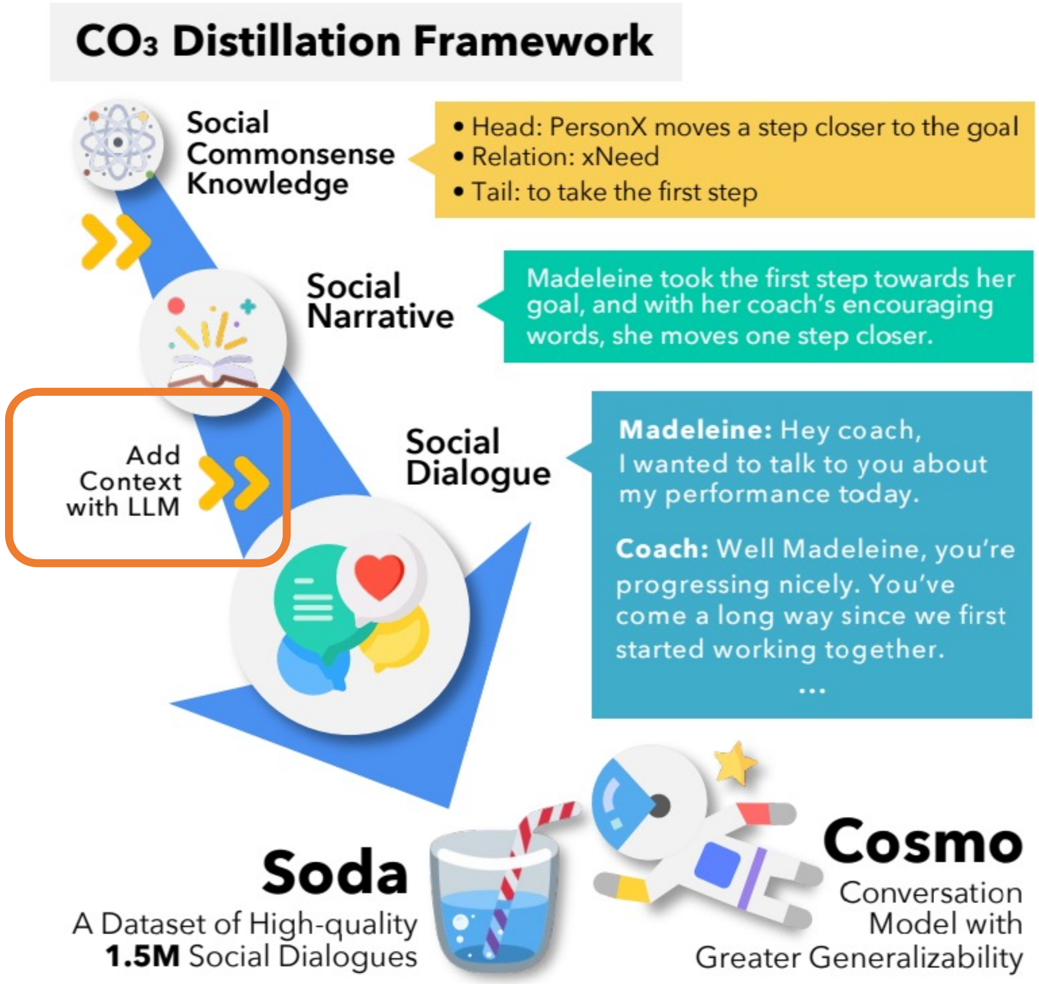
Actual utterances:
U: Exactly! I think many people fail to notice how beautiful the art of video games can be. **(PB)**
(G selected the WoW suggestion: "Indeed, Some games games are purposely designed to be a work of a persons creative expression, many though have been challenged as works of art by some critics.")
G: Indeed, Some games games are purposely designed to be a work of a persons creative expression, many though have been challenged as works of art by some critics. **(K)**
U: Video games are undervalued by many and too easily blamed for problems like obesity or violence in kids **(K)**
G: Indeed, Just last week my son was playing some Tine 2 and it was keeping him so calm. Games are therapeutic to some. **(S)**
U: I use games to relax after a stressful day, the small escape is relaxing. **(PB)**
(G selected the ED suggestion: "I enjoy doing that after a hard day at work as well. I hope it relaxes you!")
G: I enjoy a good gaming session after a hard day at work as well. **(PB)**
U: What other hobbies does your son have? **(PB)**
G: Well he likes to fly kites and collect bugs, typical hobbies for an 8 year old, lol. **(PB)**
U: My 12 year old is into sports. Football mostly. I however don;t enjoy watching him play. **(PB)**
G: I wish I could play football, But I wear this cateye glasses and they would break if I tried. **(PB)**
U: Sounds nice. Are they new or vintage? **(E)**
G: They are new, I got them because of my love for cats lol. I have to show off my beautiful green eyes somehow. **(S)**

Generating conversational data

- Example from SODA paper:
  - A method for **data generation** using large language models and knowledge graphs (named **CO_3**)
  - Data: 1,5 million written conversations in English (named **SODA**)
  - Main idea: contextualize ‘*social commonsense knowledge*’ (?)

[SODA: Million-scale Dialogue Distillation with Social Commonsense Contextualization](https://aclanthology.org/2023.emnlp-main.799) (Kim et al., EMNLP 2023)

Data generation method (CO_3)

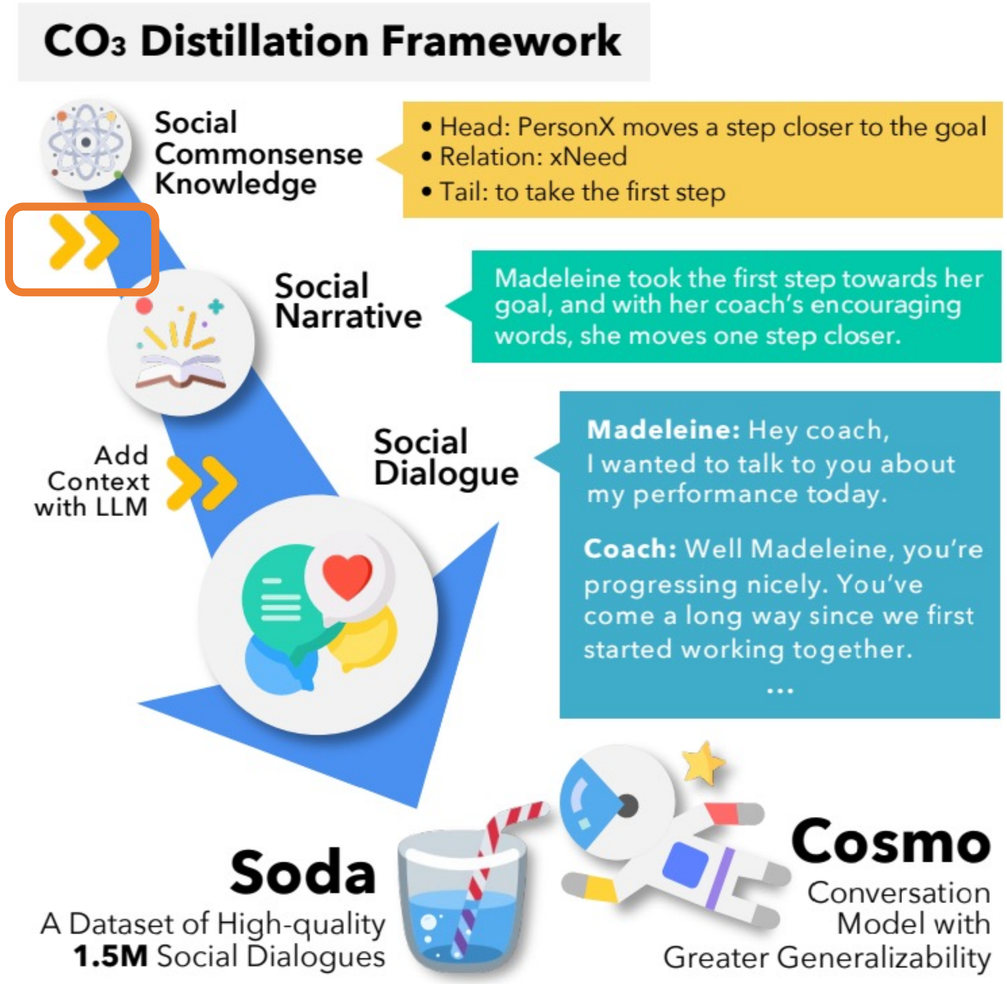


Method:

- Prompting GPT3.5 to generate/distill multi-turn written conversations
- What is defined in the prompt?
  - The speakers and the narrative of the conversation

Narrative:
Madeleine took the first step towards her goal, and with her coach's encouraging words, she moves one step closer.
Speakers: Madeleine, Coach

Data generation method (CO_3)



How is the prompt built?

- **Input:** Knowledge graph= symbolic triples
  - describing diverse **social commonsense** (Atomic10x**,** West et al., 2022)
  - A triple consists of
    - two events, denoted as the Head and Tail,
    - the relation between those two events (xNeed, xReact, etc.)
- Symbolic triples ->simple sentences
  - using templates
- Simple sentences -> final prompt
  - Again by prompting gpt 3.5

Data generation method (CO_3)

What is the **social commonsense** modelled by the knowledge graphs in Atomic10x?

- It includes emotional reactions of people to events (i.e., the xReact triples),

Event	Type of relations	Inference examples	Inference dim.
"PersonX pays PersonY a compliment"	If-Event-Then-Mental-State	PersonX wanted to be nice	xIntent
		PersonX will feel good	xReact
		PersonY will feel flattered	oReact
	If-Event-Then-Event	PersonX will want to chat with PersonY	xWant
		PersonY will smile	oEffect
		PersonY will compliment PersonX back	oWant
	If-Event-Then-Persona	PersonX is flattering	xAttr
		PersonX is caring	xAttr

Underlying assumption: social experiences form our knowledge for explaining everyday events and inferring the mental states of others

It comes from theories in psychology : Fritz Heider. 1958. The Psychology of Interpersonal Relations. Psychology Press

Data generation method (CO_3)

Post-processing the data

- Basic and commonsense filtering (manual rules to detect erroneous data)
- Safety filtering using a classifier (Canary for detecting critical conversations) and an API (Rewire: but does not seem to exist anymore) to detect toxic content
- Commonsense Filtering **prompting GPT 3.5** (again) to detect commonsense inconsistencies.

Evaluation of SODA generated content

- Pairwise comparison **with two datasets :** DailyDialog and BlendedSkillTalk (human-authored artificial conversation datasets) :
- Criteria
  - (1) which dialog has a more **natural flow?**
  - (2) context dependence: which dialogue includes responses that are more dependent on previous turns?
  - (3) **topic consistency**: which dialog is more consistent and stays on topic?
  - (4) **speaker consistency:** which dialog has speakers that are less self-contradictory ?
  - (5) **specificity**: which dialog is more **specific**
  - (6) overall

Main results

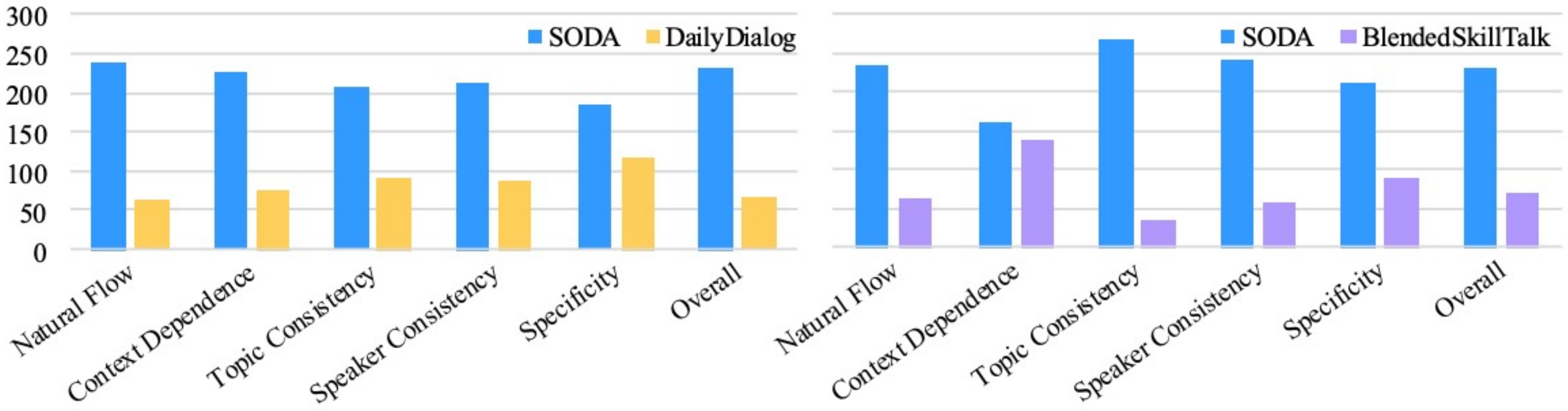


Figure 2: Results of head-to-head comparison between dialogues from SODA, DailyDialog (Li et al., 2017), and BlendedSkillTalk (Smith et al., 2020) via human judgments (§3.2). The y-axis represents the number of samples preferred by human judges. The differences in all of the categories except for the *Context Dependence* comparing SODA and BlendedSkillTalk are statistically significant ($|z| > 3.3, \quad p < 0.05$).

But … the agreement scores between workers is low (Krippendorf's alpha of 0.25, in the appendix)
The compared corpora are not so « real-life » (scripted)

Evaluating conversational systems

Warning: data contamination when evaluating systems

- The test data you use could be present in training set !
  - Pre-training data
  - Fine-tuning data
  - RLHF data
- Especially when you don’t have information about the data and task used for the model training (ex: GPT3-4)
- Some solutions have been proposed !
  - See (Shi et al., 2024) : they propose a method to compute data contamination rate.
    - based on the hypothesis that unseen data will contain more outlier words with low probability than seen data.

Shi, W., Ajith, A., Xia, M., Huang, Y., Liu, D., Blevins, T., ... & Zettlemoyer, L. Detecting pretraining data from large language models. ICLR 2024

Evaluation

- The whole framework or each module separately
  - Recognition of the user’s state
  - Prediction of the strategy of the next utterance
  - Conditional generation
- At the conversational level or at the system’s utterance level
- Ways of evaluating:
  - Automatic Measures
  - **Human evaluation**
- a challenging task for the social dimension :
  - determining what is ``appropriate'' in a conversation is rather subjective and is certainly context-dependent.

Automatic measures: at the conversation level

Measures of the **task efficiency depending on the application**

- Number of turns required to solve a task
- Learning gain for educational applications (Norman et al., 2022)
- Satisfaction gain or loss for customer relationship applications.

Norman, U., Dinkar, T., Bruno, B., & Clavel, C. (2022). Studying Alignment in a Collaborative Learning Activity via Automatic Methods: The Link Between What We Say and Do. Dialogue & Discourse, 13(2),

Automatic measures : at the system's utterance level

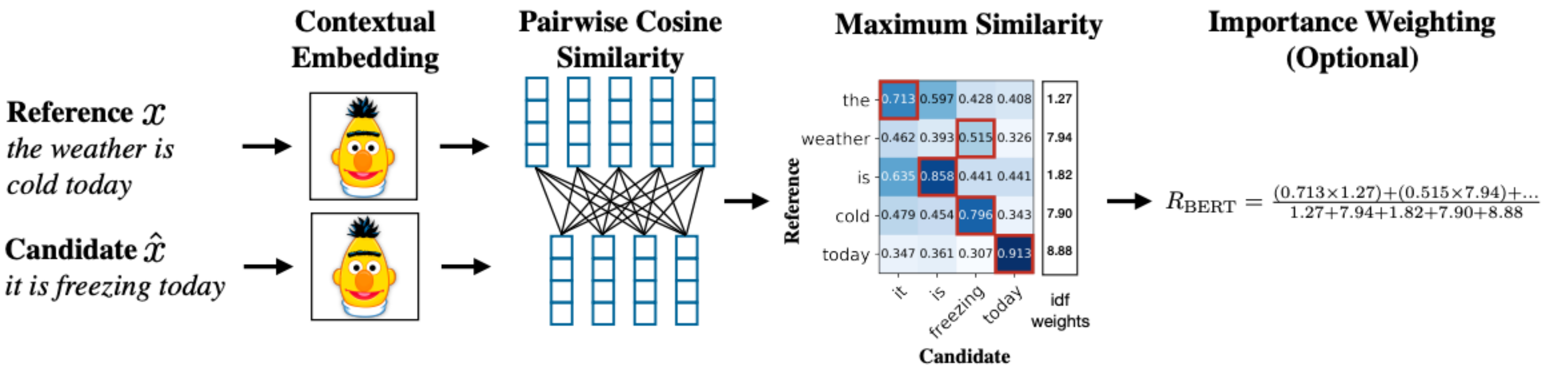
- **Reference-based :**
  - using a test dataset, compare the system's utterance s_i to the human utterance h_i for the same dialog history
  - Two types of a comparison
    - String-based
      - from 0 to 1 (exact match between si and hi)
      - cannot handle synonyms or paraphrases.
      - Ex: Bleu, Sacrebleu, Rouge, Chrf
    - Embedding-based to measure semantic similarity
      - ex. BERT Score
    - Model-based : leverage regression or pre-trained language models to return a score.
      - Ex. BART Score (the score is derived from the likelihood of regenerating the reference given the candidate text as input, or vice versa)

Automatic measures for NLG : overview

	Reference-based (Ξ)	Reference-free (¤)
String-based (§)	BLEU (Papineni et al., 2002) ROUGE (Lin, 2004) METEOR (Banerjee and Lavie, 2005) CHRF (Popović, 2015) CIDEr (Vedantam et al., 2015)	Coverage (Grusky et al., 2018) Density (Grusky et al., 2018) Compression (Grusky et al., 2018) Text length (Fabbri et al., 2021) Novelty (Fabbri et al., 2021) Repetition (Fabbri et al., 2021)
Embedding-based (ε)	ROUGE-WE (Ng and Abrecht, 2015) BERTScore (Zhang et al., 2020) MoverScore (Zhao et al., 2019) BaryScore (Colombo et al., 2021) DepthScore (Staerman et al., 2021)	SUPERT (Gao et al., 2020)
Model-based (Δ)	S3 (Peyrard et al., 2017) SummaQA (Scialom et al., 2019) InfoLM (Colombo et al., 2022b)	BLANC (Vasilyev et al., 2020)
	BARTScore (Yuan et al., 2021)	

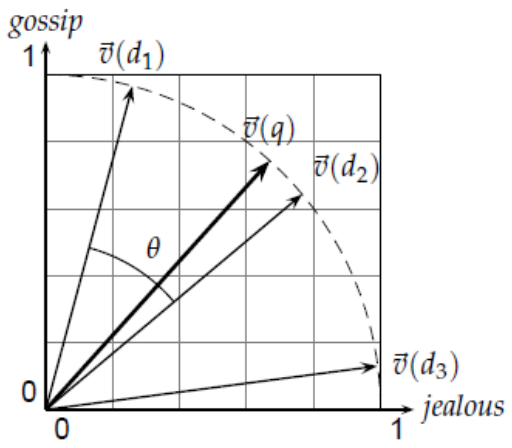
From Chhun et al., 2023

Focus on BERT Score



- Reminder (?) Cosine similarity
  - Similarity between 2 vectors d1 and d2 according to the cosine of the angle

$$\mathrm{sim}(d_1, d_2) = \frac{\vec{V}(d_1) \cdot \vec{V}(d_2)}{|\vec{V}(d_1)||\vec{V}(d_2)|},$$



► **Figure 6.10** Cosine similarity illustrated. $\mathrm{sim}(d_1, d_2) = \cos\theta$.

Automatic measures : at the system's utterance level

- **Reference-free :** measure the intrinsic quality
  - crucial because one context can yield many acceptable responses
  - at the **utterance level**, measure the intrinsic quality of each utterance
    - **Dialogue consistency**
      - Ex. BERTScore to measure the semantic distance between the generated candidate and the context history
    - **Naturalness/fluency**
      - Ex: perplexity usually used to evaluate language model but diverted here to evaluate the quality of the construction of the sentence according to a language model
    - **Social and Emotional Relevance**
      - EPITOME (Sharma et al. 2020) : a supervised model trained to predict the empathy level of a generated output
  - at the **model level**, measure the intrinsic quality of all the utterances generated by a model

Reference-free measure at the model level

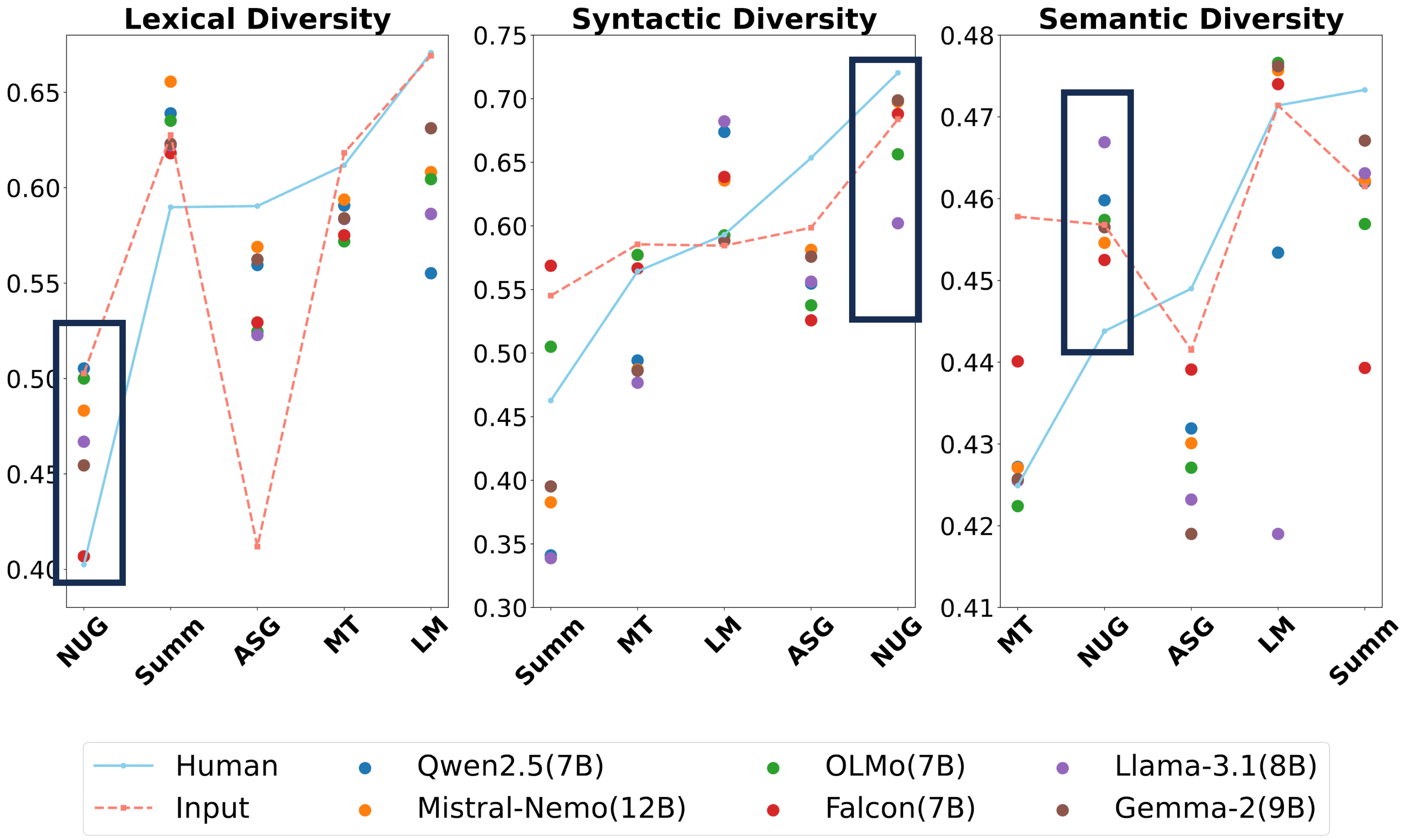
Comparing outputs of different LLMs for NUG task (Next Utterance Generation)
3 measures:

- Lexical diversity
  - Unique-n : the ration of unique n-grams to the total number of n-grams
- Semantic diversity
  - dispersion of Sentence-Bert embeddings in the semantic space
  - average pairwise cosine distance (scaled to the range [0, 1]) between all embedding vectors
- Syntactic diversity
  - Use a neural parser to generate dependency trees from sentences
  - Convert the tree into graph representations
  - Compute a distance WL between two graph representations
  - Measure the average pairwise distance

$$\boxed{\mathrm{Div}_{\mathrm{syn}}(S) = \frac{1}{\binom{n}{2}} \sum_{1 \le i < j \le n} WL(s_i, s_j).}$$

Guo, Y., Shang, G., & Clavel, C. (2024). Benchmarking linguistic diversity of large language models. arXiv preprint

Reference-free measure at the model level



Guo, Y., Shang, G., & Clavel, C. (2024). Benchmarking linguistic diversity of large language models. arXiv preprint

Human Evaluation

- At the *conversation* level or at the *system's utterance* level
- Qualitative evaluations either based on
  - a *user*'s perception
  - an *external observer*'s perception
    - annotation by crowdworkers such as in SODA paper
- Scoring/ranking/pairwise comparison of different utterances/conversations

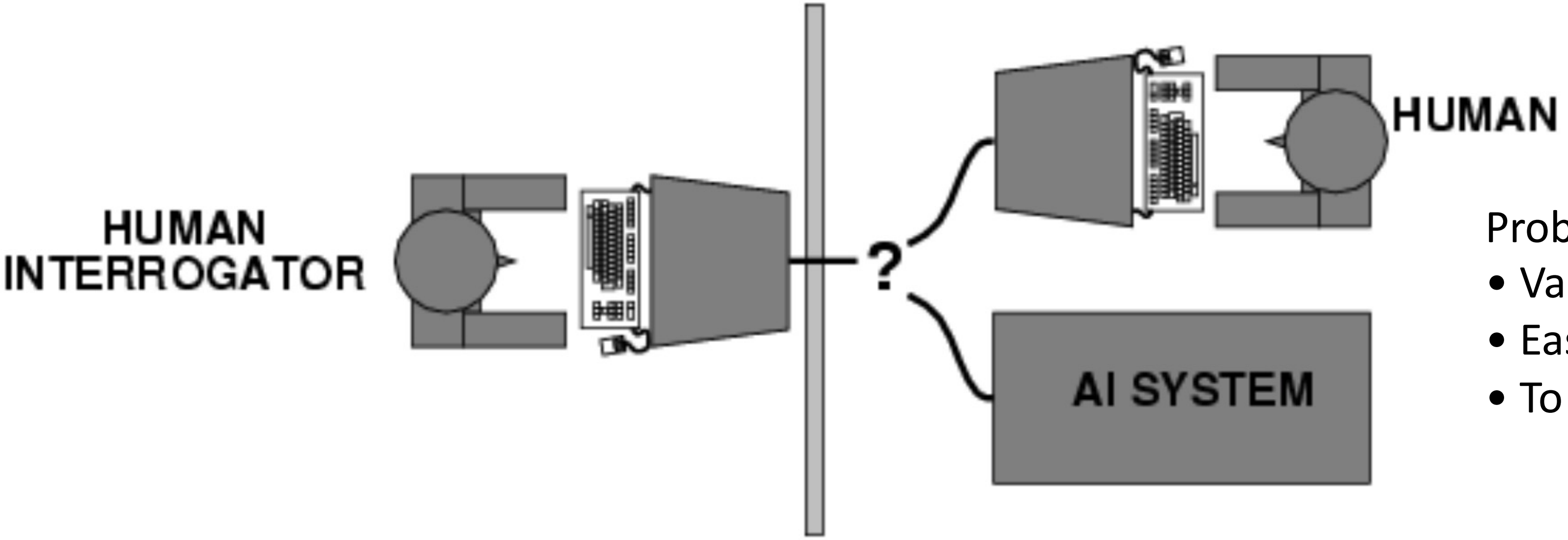
[SODA: Million-scale Dialogue Distillation with Social Commonsense Contextualization](https://aclanthology.org/2023.emnlp-main.799) (Kim et al., EMNLP 2023)

Human Evaluation

- Evaluation according to different **criteria**
    - Task-oriented, ex:
        - Ability to resolve a user’s problem for customer relationship conversational systems at the end of a conversation
        - Relevance/factuality/Specificity … of the system’s utterance
    - Socially-oriented, ex:
        - A user’s self-reported engagement at the conversation level (Sidner et al., 2005; Clavel et al., 2016)]
        - Social relevance of the system’s utterance

Popular old example for a user's perception

- The Turing test (1955) : are you interacting with a human or a system?
- Recent reformulation: a system passes the "Turing Test" if it can fool 1/3 of a group of human judges into thinking that it is a human rather than a system for at least 5 minutes
- This requires advanced conversational skills, which appears to require comprehension, reasoning and generation skills, and knowledge of the world



Problems such as
- Variability in protocols and judges
- Easy tricks can suffice (spelling mistakes)
- To pass the test means to *fool* the judges

More recent examples for a user’s perception

- Use questionnaires at the end of the conversation
  - Engagement questionnaires
    - Clavel, C., Cafaro, A., Campano, S., & Pelachaud, C. (2016). Fostering user engagement in face-to-face human-agent interactions: a survey. In Toward Robotic Socially Believable Behaving Systems-Volume II (pp. 93-120).
  - Review of existing questionnaires
    - Fitrianie, S., Bruijnes, M., Li, F., Abdulrahman, A., & Brinkman, W. P. (2022, September). The artificial-social-agent questionnaire: establishing the long and short questionnaire versions. In Proceedings of the 22nd ACM International Conference on Intelligent Virtual Agents (pp. 1-8).
  - Godspeed:
    - Bartneck, C., Croft, E., Kulic, D. & Zoghbi, S. (2009). Measurement instruments for the anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety of robots. International Journal of Social Robotics, 1(1) 71-81. | DOI: 10.1007/s12369-008-0001-3

[SODA: Million-scale Dialogue Distillation with Social Commonsense Contextualization](Kim et al., EMNLP 2023)

Example for observer’s perception

- Test on dailyDialog
  - data unseen by the evaluated model (COSMO)
- Comparison with
  - DailyDialog groundtruth
  - Outputs of pre-trained models :
    - BlenderBot
    - GODEL
    - Koala and Vicuna

Dialogue Context:
A: I really need to start eating healthier.
B: I have to start eating better too.
A: What kind of food do you usually eat?
B: I try my best to eat only fruits, vegetables, and chicken.
A: Is that really all that you eat?
B: That’s basically it.
A: How do you stick to only those foods?

Ground-truth response:
Actually, fruits and veggies are really good for you.

DialoGPT: I eat a lot of fruit and veggies. I stick to a lot of things. I don’t eat a lot of junk food.

BlenderBot: I eat a lot of fruit and veggies. I try to stay away from processed foods.

GODEL: I go to the gym and eat healthy food.

COSMO: I just try to make sure that I’m getting enough variety in my diet so that I don’t get sick of eating the same thing all the time.

Table 14: The original ground-truth response and sample responses from DialoGPT, BlenderBot, GODEL, and COSMO to a context in DailyDialog.

Example

[SODA: Million-scale Dialogue Distillation with Social Commonsense Contextualization](https://aclanthology.org/2023.emnlp-main.799) (Kim et al., EMNLP 2023)

- Comparison of pairs of *utterances* given 4 criteria:
  - Naturalness
  - Consistency:
  - Specificity
  - Overall
- The same criteria used for dialogue evaluation but here they don't distinguish between topic and speaker consistency
- Better agreement for this task (Krippendorf's alpha of 0.5)

Main results

- Largest performance gap in terms of naturalness.
- Human judges prefer COSMO's responses even over the original gold responses in the dataset

Model	Natural	Consistent	Specific	Overall
BlenderBot-3B	23%	26%	39%	28%
COSMO-3B	**77%**	**74%**	**61%**	**72%**
GODEL_L	13%	14%	15%	14%
COSMO-3B	**87%**	**86%**	**85%**	**86%**
Koala-7B	30%	34%	30%	29%
COSMO-3B	**70%**	**66%**	**70%**	**71%**
Vicuna-7B	42%	42%	44%	42%
COSMO-3B	**58%**	**58%**	**56%**	**58%**
Ground Truth	43%	45%	46%	45%
COSMO-3B	**57%**	**55%**	**54%**	**55%**

Table 5: Results of head-to-head human evaluation between model responses on an unseen dataset: DailyDialog (Li et al., 2017) (§5.1). The differences are all statistically significant with $|z| > 12.45$ and $p < 0.05$, except for the *Specific* in the bottom row.

Evaluate specific abilities of models

- The reasoning capabilities (logical or common-sense)
  - using datasets [Helwé et al., 2021, 2022]

Example 3.14

Context: David knows Mr. Zhang's friend Jack, and Jack knows David's friend Ms. Lin. Everyone of them who knows Jack has a master's degree, and everyone of them who knows Ms. Lin is from Shanghai.
Question: Who is from Shanghai and has a master's degree?
Choices: (A) David (B) Jack (C) Mr. Zhang (D) Ms. Lin

Outline

- Background
- Approaches: focus on modular approaches
- Data and evaluation
- Applications and ethical issues

Ethical issues for applications -

Potential misuses when developing conversational AI agents equipped with social and emotional capabilities:

- sway users towards
  - making purchases
  - believing misinformation

But they can also be used for good :

- Education : peer robot
- Health : pre-diagnostic assistance , help patients to find the relevant health professional

communicating and informing the users of such systems is crucial to developing their awareness of such potential risks, as well as protect them for their future interactions with AI systems.

Ethical issues for applications -

Recommendation : **Maintain a clear line between AI and humans**

It should always be clear for the user that agents are not humans (for example: self-disclosure is not an appropriate strategy)

But data used for the training is human-human interactions, models are replicating human behaviours, with a weak possibility of controlling the strategies.

Comité Ethique, Christine Noiville, Catherine Pelachaud, Patrice Debré, Jean-Gabriel Ganascia, and Raja Chatila. 2024. COMETS Opinion 2024-46 - "The phenomenon of attachment to "social robots". A call for vigilance among the scientific research community". Technical report, CNRS COMETS.

Ethical issues for methods underlying conversational systems

Reminder - What is generated by computational models and the decision-making process behind automated analyses is strongly influenced by :

- the data used to learn them (*machine learning*)
  - Ex : GPT3 has been learned from 300 billion tokens (~words) extracted from Wikipedia, books, and crawled pages on the web
  - human annotations used to supervise (guide) the learning process, ex:
    - Annotation of opinions expressed in texts
    - Annotation of the "ethical" or "safe" character of the answer
    - For chatGPT: redaction of standard responses by humans

But: the definition and coding of what constitutes the social norm is not only task-dependent but also culture-dependent and may rightly be controversial (Ferrari et al., 2016).

Ethical issues for methods underlying conversational systems

- In modular approaches the dialog policy of the system is explicit which is not the case in end-to-end approaches
- But now end-to-end approaches are preferred
- -> lost of transparency

Take home messages

1. Natural dialogue in natural language is still an on-going challenge, implying lot of links with several research disciplines: linguistics, pragmatics, cognitive sciences, etc.
2. Neural models have increased the performance of conversational systems, but classical symbolic techniques are still relevant, and the future may be hybrid to keep a sufficient level of explainability …
3. Be aware of the encoded subjectivity and on ethical aspects

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